

What Held-Out Scores Predict About Deploying Intrusion Detection in O-RAN

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ABSTRACT Several 5G and O-RAN intrusion detection papers report more than 99% accuracy on a random split of one corpus. Earlier studies measured transfer or latency separately. We measure transfer, alert burden, and latency for six architectures and two input-blind baselines on three public corpora (NetsLab-5GORAN-IDD, 5G-NIDD, and a 5G core testbed). On the radio layer, a random split raises macro- F_1 by 0.13 over a session-disjoint split (95% interval -0.01 to 0.27), and a nearest-neighbor lookup gains 0.14, consistent with recognition of capture sessions. In 5G-NIDD, 59% of benign flows are copies of flood records labeled as attacks, and their addresses point to the attacker at the other base station. The copies cap the balanced accuracy of any content-only classifier at 0.766. Two record-position fields tell them apart: without these fields the reproduced published pipeline falls from 99.9% to 76.9% accuracy, yet it stays near 100% on flows without the copies. On flows with consistent labels, detectors transferred to 5G-NIDD reach 0.61 to 0.78 balanced accuracy, below in-target references (0.842 to 0.982), with a loss against held-out source data that is significant for one of six architectures. A common exporter does not close the transfer gap. At a declared prevalence of 0.002, no transferred detector exceeds a precision of 0.092 at recall above 10%, and none passes our deployability test. Window-level detectors meet its latency term in a FlexRIC loop with an emulated E2 node (99th percentile at most 4.95 ms). A random-split score is not evidence of deployability.

INDEX TERMS Dataset shift, evaluation methodology, false positive rate, intrusion detection, label quality, O-RAN, RAN intelligent controller, xApp.

I. INTRODUCTION

Open Radio Access Network (O-RAN) splits the base station into radio, distributed, and central units joined by open interfaces and adds a RAN Intelligent Controller (RIC) that runs third-party control applications [1]–[3]. The near-real-time RIC sits close to the cells it controls [1], [4] and receives the per-device measurements that their E2 nodes report under its subscriptions. A detection application (xApp) in the RIC could therefore look for attacks close to where they enter the network, for example from compromised IoT devices, a well-documented source of volumetric attacks [5], [6]. Platforms for building xApps exist [7], [8], and detectors in or beside the O-RAN control loop have been built and evaluated, each in its own testbed [9], [10].

Several studies of machine-learning intrusion detection for 5G evaluate on a random held-out split of one dataset and report accuracy or F_1 above 99%; on 5G-NIDD, one of our three corpora, the dataset authors report binary accuracies of 99.85% to 99.95% for four of five classifiers [11], and later work a weighted F_1 of up to 99.95% [12]. A random split measures how well a detector separates the records of

one corpus, not the three properties that deployment depends on: generalization to a site that contributed no training data, where intrusion detectors are known to transfer poorly [13], [14]; alert burden, since a false positive rate that looks small on a balanced benchmark becomes a stream of alerts at production flow rates and a realistic attack base rate [15], [16]; and latency, since the O-RAN E2 specification allows near-real-time control loops of 10 ms to 1 s [17]. We ask what a held-out score on one corpus predicts about these properties, and, unable to observe a deployment, study generalization through transfer to independently collected corpora.

For NetsLab-5GORAN-IDD and 5G-NIDD, two of our corpora, Abraheem and Edhirig measured transfer [18] and Obiuevwi et al. timed inference inside a real near-real-time RIC [19]. We offer neither a new detector nor the finding that transfer hurts, but measure random-split leakage, transfer loss, alert burden, and latency for the same six architectures, with input-blind baselines on every corpus, so that the sizes of these effects can be compared. Doing so exposed a labeling fault: half of 5G-NIDD lies on records that occur under both

labels, and a published 99.9% on it depends on two fields that encode a record's position in its file. Our contributions are:

- A ladder of evaluation protocols, from a random split to splits that hold out capture sessions, capture files, base stations, and whole corpora. On the radio layer the random-split gain is consistent with the recognition of capture sessions, and the capture schedule rules out a drift estimate (Sections VI-A and VI-B).
- An audit of the 5G-NIDD labels that traces the benign-labeled flood copies to the attacker at the other base station, and the published pipeline moved down the same ladder (Section VI-C).
- Separate tests of what the gap between corpora consists of: prior shift, label conflicts, the flow exporter, and the traffic itself (a domain classifier and a third corpus) (Sections VI-D to VI-F).
- Operational precision and alert volume from pooled confusion counts with cluster-bootstrap intervals, where a mean of per-split precision would inflate differences between architectures by an order of magnitude (Section VI-G).
- Which calibration methods can change the reachable operating points; none raises target balanced accuracy here, and label-free estimates of the target prior fail once the class-conditional distributions shift (Section VI-H).
- Decision latency on one host for three implementations, and the full loop of a window-level detector running as an xApp in FlexRIC with an emulated E2 node (Section VI-I).
- A deployability test that combines generalization, alert burden, and latency, its sensitivity to its bounds, and reporting requirements (Sections VII and X).

Two limits of scope apply throughout. We measured latency on one host, without a RIC and through FlexRIC with an emulated E2 node, with no radio or operator network. The three corpora also come from three flow exporters, so every cross-corpus gap mixes a change of site with a change of tooling; for $\mathcal{D}_A$ and $\mathcal{D}_B$ we reduce this confound by re-extracting $\mathcal{D}_B$ with Zeek, the exporter family of $\mathcal{D}_A$, whose configuration for $\mathcal{D}_A$ is undocumented (Sections VI-E and IX).

II. BACKGROUND AND RELATED WORK

A. THE NEAR-REAL-TIME RIC AS A DETECTION PLATFORM

An xApp talks to E2 nodes through E2 service models (E2SM). E2SM-KPM delivers periodic key performance measurements (KPM), and E2SM-RC exposes control actions [1]. Platforms such as OpenRAN Gym [20], SCOPE [21], CoO-RAN [7], and FlexRIC [8], with open radio stacks [22], [23], made experimental xApps practical. E2 carries aggregated measurements, not packets, so a flow-level detector needs a data path of its own (Section VII-A).

B. SECURITY OF OPEN RAN

The RIC and its third-party xApps are themselves a new attack surface, according to the O-RAN Alliance threat

model [24] and independent analyses [25], [26], and security mechanisms for the O-RAN interfaces have since been built and evaluated [27]. On the detection side, 5G-Spector offers layer-3 attack detection as an O-RAN compliant service [9], Det-RAN detects attacks from cross-layer data in real time [10], and Fard et al. fuse radio telemetry and flow records on our source corpus [28]. Each is evaluated in the setting it was built for; our question is what such an evaluation predicts about another site.

C. MACHINE LEARNING FOR NETWORK INTRUSION DETECTION

Public intrusion corpora are plentiful for IoT traffic [29]–[33] and scarce for cellular networks. 5G-NIDD was captured on an operational 5G test network [34], [35], and NetsLab-5GORAN-IDD pairs flow records with radio telemetry from an O-RAN testbed [36]. Standardized NetFlow feature sets made cross-corpus experiments easier [37], [38]. The labels and flow records of a corpus can themselves be wrong: Engelen et al. documented labeling and flow-construction faults in CIC-IDS2017 [39], and Flood et al. found design flaws across NIDS benchmarks that let a classifier score well without learning the attack [40].

D. EVALUATION PRACTICE AND DATASET SHIFT

Reported results tend to overstate what a detector can do, for well-known reasons: closed-world evaluation [16], neglect of the base rate [15], sampling bias and data leakage [41], [42], and temporal or spatial bias in the split [43]. Apruzzese et al. trace the gap between reported and operational effectiveness across security domains [44], Kus et al. find it in industrial intrusion detection once detectors meet data they were not tuned on [45], and cross-dataset generalization has repeatedly proved poor [13], [14], [46].

Closest to our work, Abraheem and Edhirig transfer XGBoost between the same two corpora with 15 harmonized features and report zero-shot balanced accuracies per attack family of 0.50 to 0.56 from NetsLab-5GORAN-IDD to 5G-NIDD and 0.52 to 0.83 in the other direction [18]. We add input-blind baselines, in-target references, the published pipeline under the same protocol ladder, an audit of the target labels, and the operational and latency criteria.

From the dataset-shift literature we take label-shift estimation from unlabeled data [47], [48], which assumes that only the class balance changes, calibration and its behavior under shift [49]–[51], shift detection [52], and covariance alignment [53], which matches the second moments of the source features to the target. Classifiers compared under repeated resampling are tested as Nadeau and Bengio recommend [54].

III. DEPLOYMENT CRITERIA AND THREAT MODEL

A. SYSTEM AND THREAT MODEL

A near-real-time RIC serves a set of E2 nodes (Fig. 1) to which user devices, IoT devices among them, attach. The adversary controls some devices and uses them for flooding, scanning, brute-force login, web attacks, or, in one corpus, an

attack on the control plane of the core network (PCF session deletion), the attack categories in our corpora. It does not control the RIC, the E2 interface, or the xApp. The defender's xApp builds radio telemetry windows from E2SM-KPM reports or reads flow records from a user-plane exporter outside the RIC (Section VII-A), and it can act through E2SM-RC.

B. CRITERION 1: GENERALIZATION ACROSS CORPORA

Let $\mathcal{D}_A$ and $\mathcal{D}_B$ be two independently collected corpora in a shared feature space. We measure a model f_θ trained on $\mathcal{D}_A$ by how far its score on $\mathcal{D}_B$ falls below its held-out score on $\mathcal{D}_A$. The corpora differ in attack prevalence (Table 1), and a change in macro- F_1 blends a change in the classifier with a change in class balance: even a classifier that ignores its input scores differently when the prevalence moves. We therefore use the difference in balanced accuracy (BA, the mean of the true positive and true negative rates), which prevalence does not affect,

$$\Delta_{BA} = \text{BA}(f_\theta; \mathcal{D}_A^{\text{test}}) - \text{BA}(f_\theta; \mathcal{D}_B), \quad (1)$$

and we also give the macro- F_1 gap Δ_{F_1} , defined in the same way (source minus target), and its difference from the gap of the majority-class baseline. As a descriptive diagnostic that is not part of the test, we measure covariate shift one feature at a time,

$$\bar{W}_1 = \frac{1}{|\mathcal{F}|} \sum_{j \in \mathcal{F}} W_1(Q_j(P_A^{(j)}), Q_j(P_B^{(j)})), \quad (2)$$

where $\mathcal{F}$ is the set of 18 shared columns, $P_A^{(j)}$ and $P_B^{(j)}$ are the marginal distributions of column j in $\mathcal{D}_A$ and $\mathcal{D}_B$, W_1 is the first-order Wasserstein distance, and Q_j is the quantile transform of column j , fitted on source training data only. Had each corpus been normalized by its own quantiles, both marginals would become uniform and every term would vanish.

C. CRITERION 2: ALERT BURDEN UNDER A REALISTIC BASE RATE

Let λ_b be the benign arrival rate in the detector's own unit, which is flow records per hour for a flow-level detector and telemetry windows per hour for a window-level one. A benign user equipment (UE) produces 225 non-overlapping windows of 16 s per hour. For a decision threshold τ on the detector's score, let $\text{FPR}(\tau)$ be the false positive rate and $R(\tau)$ the recall. False alerts then arrive at

$$A(\tau) = \lambda_b \text{FPR}(\tau), \quad (3)$$

and, for an operational attack prevalence π , the fraction of alerts that correspond to attacks is

$$\text{PPV}(\tau) = \frac{\pi R(\tau)}{\pi R(\tau) + (1 - \pi) \text{FPR}(\tau)}. \quad (4)$$

We call this PPV the operational precision; the corpus precision is the precision at the corpus's own attack prevalence. The corpora cannot give us π or λ_b , so we declare both (Section V). Only π enters the PPV, and A grows linearly with

λ_b , so alert counts at another λ_b follow from the false positive rates we report.

D. CRITERION 3: LATENCY AGAINST THE CONTROL BUDGET

The decision latency of an xApp can be decomposed as

$$L = t_{\text{ind}} + t_{\text{feat}} + t_{\text{inf}} + t_{\text{act}} + t_q, \quad (5)$$

where the terms are the delivery of the indication to the xApp, feature construction, inference, the control action up to its acknowledgment, and queueing inside the xApp. We time feature construction and inference alone on one host, and all five terms for window-level detectors in a loop through a near-real-time RIC with an emulated E2 node (Section V). What matters for conformance is the tail: for a budget B and a tolerated violation rate ε , we require $q_{1-\varepsilon}(L) \leq B$. The test of Section III-E uses the strictest budget the E2 specification allows, $B = 10$ ms, and we also report the smallest budget $B_{\min}$ each detector meets.

E. A DEPLOYABILITY TEST

The three criteria are combined into a test that a detector either passes or fails:

$$\begin{aligned} \mathcal{P}(f_\theta) = & \mathbf{1}[\bar{\Delta}_{BA} \leq \delta] \cdot \mathbf{1}[\bar{q}_{0.99}(L) \leq B] \\ & \cdot \mathbf{1}[\exists \tau : R(\tau) \geq r_{\min}, A(\tau) \leq A_{\max}, \\ & \text{PPV}(\tau) \geq \rho], \end{aligned} \quad (6)$$

where $\bar{\Delta}_{BA}$ and $\bar{q}_{0.99}$ are upper 95% confidence bounds, so a pass cannot hinge on a lucky point estimate. The generalization and alert terms are evaluated on the same target flows; the latency term needs a measured decision loop, which we have for window-level detectors only. The minimum recall $r_{\min}$ keeps a detector that never raises an alert from passing the alert term, and a passing detector would operate at the most sensitive admissible threshold, the one with the highest recall among those meeting all three alert conditions. The test is a reporting instrument, not an O-RAN requirement; its bounds stand for operator requirements, and Section VII-B gives our values and how the verdict responds when they change.

IV. METHODOLOGY

A. CORPORA

$\mathcal{D}_A$ is NetsLab-5GORAN-IDD [36], recorded on a physical OpenAirInterface O-RAN testbed, with benign traffic and five attack categories: denial of service (DoS), distributed denial of service (DDoS), probing, brute force, and web attacks. Its two layers, flow records exported by Zeek at the central unit and radio telemetry sampled at 1 Hz at the distributed unit, cannot be joined record by record from the published files. For the radio layer we cut the records of each UE into consecutive, non-overlapping windows of 16 records and summarize each window by the mean and standard deviation of 16 KPM fields.

Gaps of more than five minutes in the timestamps split the radio layer into 30 capture sessions, the group key of every

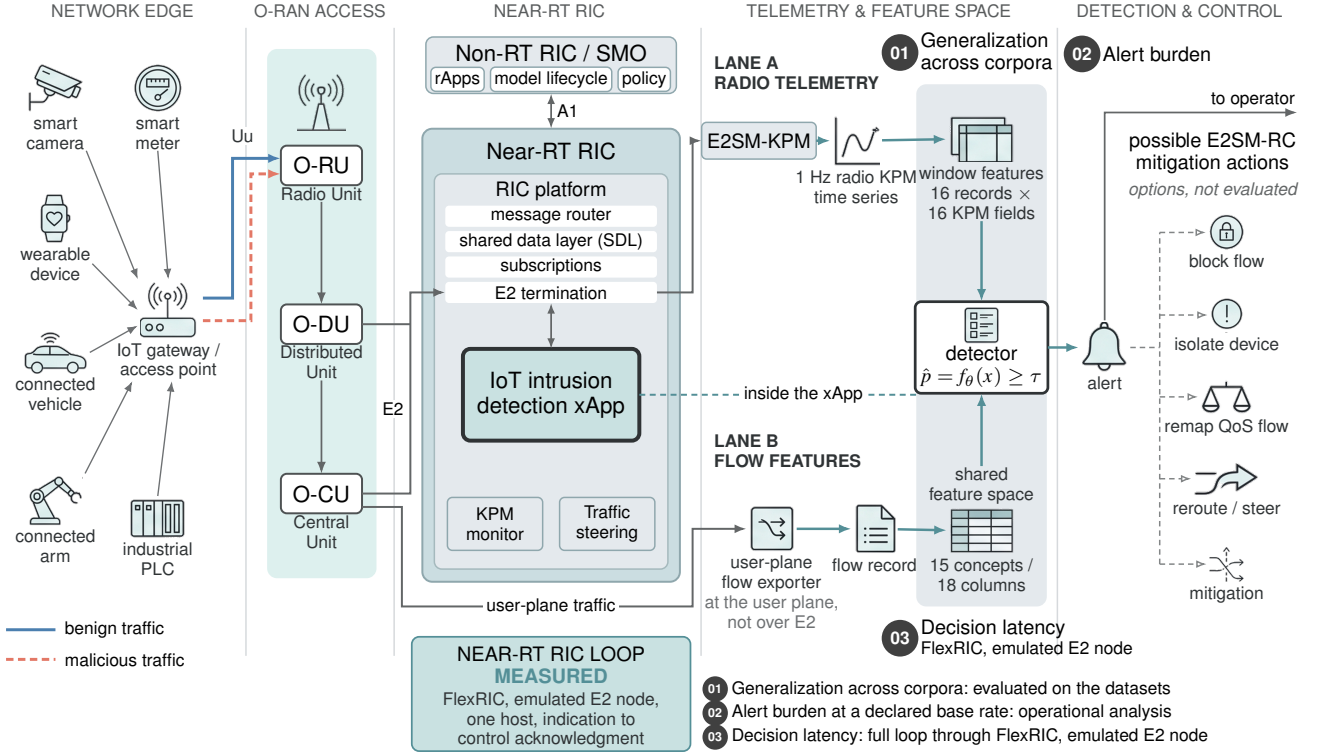


FIGURE 1. Reference O-RAN edge deployment and the detection loop. The detection xApp receives KPM reports through the E2 termination (lane A). Lane B assumes that flow records from a user-plane exporter reach the xApp, a path E2 does not provide (Section VII-A), and maps them onto the shared feature space. The telemetry column expands the xApp’s feature construction and inference, one detector per layer. Numbered markers show what we measure: generalization across corpora (01) and alert burden (02) on the corpora, and decision latency (03) for window-level detectors, from indication to control acknowledgment, through FlexRIC with an emulated E2 node on one host. The mitigation actions on the right are not evaluated.

radio split. Fard et al. describe the corpus as 42 experiment runs [28], and a gap threshold of 90 to 105 s recovers exactly 42 segments, each with a single attack category. Every session is a union of complete segments: the five-minute rule merges runs recorded back to back, keeping them on the same side of any split, and 10 of the 30 sessions contain two or three attack subcategories, the finer attack labels the corpus gives within a category. All records of a session carry the same category label, so a radio label is a session label. Section VI-B gives the capture schedule.

$\mathcal{D}_B$ is 5G-NIDD [34], [35], captured at another institution on a 5G test network with two base stations, exported by Argus, and labeled with two attack categories (DoS and probing). Its combined file, *Combined.csv*, has no addresses, ports, or timestamps but keeps the Argus record counter *Seq* and the file offset *Offset*, which restarts at every file boundary, so we can recover 20 capture files, ten per base station (files 1 to 10 from station 1, 11 to 20 from station 2). Both stations went through the same sequence of attack types: nine files each hold benign traffic plus one attack type, and the tenth carries no attack label. A second release, which we call the full-field release, keeps every Argus field, addresses included, and the raw captures are public too.

A third corpus, $\mathcal{D}_C$, gives a second transfer target at a third site: the flow records of a 5G core testbed (Open5GS in Docker) published by Nugraha et al. [55]. Three of its files

TABLE 1. Datasets. $\mathcal{D}_A$ supplies training and in-distribution testing on both layers. $\mathcal{D}_B$ and $\mathcal{D}_C$ are transfer targets, and the in-target references of each are trained and tested on it. $\mathcal{D}_B$ is also the training source for the reverse direction and for transfer to $\mathcal{D}_C$.

	$\mathcal{D}_A$ flows	$\mathcal{D}_A$ radio	$\mathcal{D}_B$	$\mathcal{D}_C$
Unit	flow	16 s UE window	flow	flow
Source	Zeek	KPM, 1 Hz	Argus	NFStream
Samples	1 640 182	2 808	1 215 889	39 425
Features	18	32	18	18
Attack prevalence	94.63%	76.46%	60.71%	2.13%
Attack categories	5	5	2	3
Groups	318 hosts	30 sessions	20 files	3 files
Role	train, test	train, test	target	target

hold flow records, each with one attack (a SYN flood, an ICMP flood, and a PFCP session-deletion attack), for 39,425 flows, 839 of them attacks; its benign flows mix user traffic with core-network signaling. NFStream, which exported $\mathcal{D}_C$, counts link-layer bytes plus 44 bytes of GTP-U encapsulation per tunneled packet, and we strip both so that the shared columns hold inner IP bytes, as for $\mathcal{D}_A$. Table 1 summarizes the datasets.

B. SHARED FEATURE SPACE

Zeek and Argus share no column names, so we define the flow feature space by meaning (Table 2): 15 concepts in 18 columns, namely seven base quantities (duration and

TABLE 2. Shared flow feature space, generated from `configs/features/shared_space.yaml`. The exporter-robust subset is the protocol one-hot, the three mean packet sizes, and the two direction ratios.

Concept	$\mathcal{D}_A$ (Zeek)	$\mathcal{D}_B$ (Argus)	Role
duration	duration	Dur	base
src_bytes	src_ip_bytes	SrcBytes	base
dst_bytes	dst_ip_bytes	DstBytes	base
src_pkts	src_pkts	SrcPkts	base
dst_pkts	dst_pkts	DstPkts	base
tot_pkts	src_pkts + dst_pkts	TotPkts	base
tot_bytes	src_ip_bytes + dst_ip_bytes	TotBytes	base
proto	proto	Proto	one-hot (4)
mean_pkt_size	tot_bytes / max(tot_pkts, 1)		derived
src_mean_pkt_size	src_bytes / max(src_pkts, 1)		derived
dst_mean_pkt_size	dst_bytes / max(dst_pkts, 1)		derived
bytes_per_s	tot_bytes / max(duration, duration_floor_s)		derived
pkts_per_s	tot_pkts / max(duration, duration_floor_s)		derived
src_byte_ratio	src_bytes / max(tot_bytes, 1)		derived
src_pkt_ratio	src_pkts / max(tot_pkts, 1)		derived

packet and byte counts), a one-hot protocol with a fixed list of levels, and seven derived quantities computed the same way for every corpus. One mapping is easy to get wrong. Zeek’s `src_bytes` counts application payload, zero for most floods and scans, whereas Argus’s `SrcBytes` counts whole packets including link-layer framing; its counterpart is Zeek’s `src_ip_bytes`, which counts a minimum-size packet as 40 bytes where Argus counts 42. For $\mathcal{D}_C$, NF-Stream’s duration, per-direction packet and byte counts (without the overheads above), and protocol number fill the same columns. All corpora get the same 1 ms floor on duration and $\log(1 + x)$ transform for unbounded quantities, scaling is fitted on source training data alone, and no address, port, or timestamp is a feature. Projecting $\mathcal{D}_A$ onto the shared columns costs 0.005 to 0.082 of in-distribution balanced accuracy against all 46 non-port Zeek features.

Before computing any result with it, we also fixed a subset of nine columns that we expected to depend less on the exporter, which we call exporter-robust: the protocol and the five per-packet means and direction ratios. Where an exporter ends a flow moves duration, rates, and totals; per-packet means and direction ratios move less but are not invariant, since splitting a flow changes them and the 40- and 42-byte counts differ by about 5% for minimum-size packets.

C. DETECTORS

Every protocol is run on six supervised architectures, logistic regression (LR), a decision tree (DT) of depth 12, a random forest (RF) [56], XGBoost (XGB) [57], histogram gradient boosting (HGB) from scikit-learn [58], and a multilayer perceptron (MLP), and on two input-blind baselines, a majority-class predictor and a stratified sampler of the training prior, whose scores can change between protocols only through class composition. All supervised models use balanced class weights (sample weights for the MLP, `scale_pos_weight` for XGB). We fixed the hyperparameters in advance, because tuning on held-out source data would reward source-specific structure, which the transfer

experiments measure, and tuning on the target would leak target information into training. LR uses `max_iter` 2000 on standardized inputs, RF 200 trees with minimum leaf size 2, XGB 300 trees of depth 6 with $\eta = 0.1$, subsample 0.8, and the `hist` method, HGB 300 iterations of depth 6 with $\eta = 0.1$, and the MLP two hidden layers of 128 and 64 units trained with Adam and early stopping (patience 10) on standardized inputs.

Three more model families answer narrower questions. Two sequence models, a gated recurrent unit (GRU) network [59] with 32 hidden units and attack-class weight equal to the benign-to-attack ratio, and a one-dimensional convolutional network (1D-CNN) with two convolutions of 32 filters, kernel 3, and average pooling, read the raw 16-step KPM sequences of the same radio windows. Two unsupervised detectors, an isolation forest (IF) [60] with 200 trees and an autoencoder (AE) with a 12–6–12 layout scored by reconstruction error, learn from benign source flows only, with the alarm threshold at the 99th percentile of benign source scores. A CORAL variant [53], the one transfer model that sees target data, re-colors the source flows to the covariance of the unlabeled target before training (per-domain z-scores, identity-regularized covariances).

Two analyses use no detector: a one-nearest-neighbor (1-NN) lookup that gives each radio test window the label of its nearest training window (Euclidean distance on standardized window summaries), and a domain classifier (histogram gradient boosting, 200 iterations) that tells $\mathcal{D}_A$ from $\mathcal{D}_B$ flows, scored by balanced accuracy under five-fold stratified cross-validation.

D. EVALUATION PROTOCOLS

Splits are group-disjoint wherever a group key exists, apart from random splits run as a comparison, and an automated check rejects any group-disjoint split with a group on both sides. On the radio layer the group key is the capture session, and three protocols each hold out about 20% of the windows: a random split, a session-disjoint split, and a category-stratified session-disjoint split that keeps sessions of every attack category on both sides. Two further designs probe time (Section VI-B): holding out the latest or earliest session of every category, against control draws that hold out a random session per category, and holding out each benign session alone. On the flow layer the group key is the source address (318 groups). For each of 20 split seeds we draw a seeded sample of 300,000 flows from $\mathcal{D}_A$ (a size fixed before the first run), hold out about 20% of it by source address, and evaluate on all of $\mathcal{D}_B$. The reverse direction trains on 80% of a 300,000-flow sample of $\mathcal{D}_B$ and tests on $\mathcal{D}_A$. The in-target references on $\mathcal{D}_B$ train on 80% of such a sample and test on the rest, hold out four of the 20 capture files, or train on one base station and test on the other (Section VI-C). Apart from these references, CORAL, the domain classifier, the prior estimates of Section VI-H, and the best-threshold searches reported as upper bounds, no transform, threshold, or model choice touches $\mathcal{D}_B$, and every access to it is logged.

E. STATISTICAL ANALYSIS

We estimate uncertainty over 20 splits, each defined by its split seed. The splits resample one population of 30 sessions or 318 addresses, so the training sets of different splits overlap, as do their test sets, and a paired t -test over split means understates the variance. Our intervals are therefore Nadeau–Bengio corrected [54]: the variance is scaled by $1/J + n_{\text{test}}/n_{\text{train}}$ for J splits, with n counting the windows or flows of the source test and training sets. In a transfer comparison the target is fixed, so the interval covers only the variation from resampling the source, not that across target corpora. Holm correction is applied within each family, one metric and one comparison (a protocol pair or a transfer direction) across the six supervised architectures; the two sequence models form a family of their own. Operating points come from confusion counts pooled over splits (Section VI-G). A pooled count includes a sample once per split that tests it, and samples in one cluster are correlated, so instead of a binomial interval we use a cluster bootstrap with 2,000 replicates that resamples both splits and clusters (radio sessions, $\mathcal{D}_A$ source addresses, or $\mathcal{D}_B$ capture files).

F. DECISION LOOP OF THE XAPP

Algorithm 1 gives the decision loop of a detection xApp, which appends each KPM report to the window of its UE and scores the window once it holds 16 records, one decision per UE every 16 s. The deadline check compares the time ℓ spent inside the xApp with an in-xApp budget B_x : B minus the time the operator reserves for delivering the indication (t_{ind}) and acknowledging the control message (part of t_{act}). A late decision is still acted on and logged as a violation, assuming that a late mitigation can still help; to drop late actions instead, the check would move before the transmission. Our timing xApp (Section V) differs in three ways: at every report it scores a held-out test window for every UE, as if each report completed a window; it sends a control message after every report, so that t_{act} is always measured; and it records the full loop instead of applying the check.

V. EXPERIMENTAL SETUP

All measurements ran on one workstation (13th Gen Intel Core i9-13900H, 14 cores, 20 threads, 47.6 GB of memory, Windows 11 Pro build 26120) with Python 3.11.9, scikit-learn 1.8.0, XGBoost 3.2.0, NumPy 2.3.5, and ONNX Runtime 1.30.0; the sequence models were trained with PyTorch 2.14 on the CPU in a separate environment. The full package list ships with the results.

For the latency loop we built FlexRIC [8] v2.0.0 in WSL2 (Ubuntu, Linux kernel 6.18) on the same machine and connected its near-real-time RIC over SCTP to FlexRIC's emulated gNB agent. Our C xApp subscribes to E2SM-KPM reports with 16 measurement items per UE (one radio record), scores every UE of a report with ONNX Runtime 1.30.0 through its C API, answers each report with one E2SM-RC control message (radio bearer control, QoS flow mapping, the only control the emulated node implements), and waits for the

Algorithm 1 Budget-aware decision loop of a detection xApp

Require: indication stream $\mathcal{S}$, model f_θ , threshold τ , in-xApp budget $B_x < B$

Ensure: E2 control actions; violation count v ; latency record

```

1:  $\mathcal{W} \leftarrow$  empty per-UE window map;  $v \leftarrow 0$ 
2: for all indication  $m$  arriving on  $\mathcal{S}$  do
3:    $t_0 \leftarrow \text{Now}()$ ;  $u \leftarrow \text{UEOf}(m)$ 
4:    $\mathcal{W}[u] \leftarrow \text{Append}(\mathcal{W}[u], m)$ 
5:   if  $|\mathcal{W}[u]| = 16$  then  $\triangleright$  non-overlapping windows
6:      $x \leftarrow \text{Features}(\mathcal{W}[u])$ ;  $\mathcal{W}[u] \leftarrow$  empty
7:      $\hat{p} \leftarrow f_\theta(x)$ 
8:     if  $\hat{p} \geq \tau$  then
9:        $\text{SendE2}(\text{EncodeControl}(u, \hat{p}))$   $\triangleright$  even if late
10:    end if
11:     $\ell \leftarrow \text{Now}() - t_0$ 
12:    if  $\ell > B_x$  then
13:       $v \leftarrow v + 1$   $\triangleright$  deadline violation, recorded
14:    end if
15:     $\text{RecordLatency}(\ell)$   $\triangleright$  every decision
16:  end if
17: end for
```

acknowledgment. The emulated node reports synthetic values for 0 to 12 UEs, so the xApp scores held-out test windows in their place; before timing, the C code reproduced the Python scores on all 589 test windows to within 4×10^{-7} . Delivery time comes from the timestamp the node writes into the KPM header, since all processes share one clock. No radio, real E2 node, or container platform took part, and all other latency figures were measured without a RIC.

The radio protocol comparison uses 20 splits with two model seeds each, and the other experiments 20 splits with one. For the flow-level bootstrap intervals and every result on $\mathcal{D}_B$ without the benign copies, we re-ran the first 10 splits deterministically with confusion counts per capture file (runtime limited the re-run to 10). Calibration uses 10 splits of a three-way source split into training, calibration, and test groups. We declare an attack prevalence of $\pi = 0.002$ and a benign arrival rate of $\lambda_b = 240,000$ flows per hour. Neither comes from an operator, and we found no measured source for π ; Section VII-B gives the prevalence at which the precision conclusion would change. On the radio layer, false alerts are counted per benign UE-hour. Our rate of about 67 flows per second is below the several hundred per second that Jin et al. extrapolate from measured radio-bearer arrivals at a heavily loaded base station of a large LTE network [61], so our alert volumes probably understate those of an edge site serving several cells.

Latency outside the RIC is measured per call with a monotonic clock and garbage collection enabled, over 20,000 calls after 1,000 warm-up calls, with one thread per library unless stated otherwise, and every quantile has a distribution-free confidence interval. To time flow-level feature construction, we use two Python flow exporters of our own with the same flow semantics, a per-packet reference implementation and a

vectorized version that extracts header fields in bulk, on the three DDoS captures of $\mathcal{D}_A$.

VI. RESULTS

A. EVALUATION-PROTOCOL SENSITIVITY

Table 3 and Fig. 2 compare three protocols on the radio layer. Against a session-disjoint split, a random split raises macro- F_1 by 0.09 to 0.17 across the six architectures, 0.13 on average, and the gain is positive in at least 16 of the 20 splits for every architecture. The input-blind baselines gain nothing, so class composition alone does not move macro- F_1 between protocols, which is why we compare protocols in macro- F_1 on this layer. The corrected interval of the average gain is $[-0.01, 0.27]$ ($p = 0.06$): with thirty sessions we cannot pin down the size of the effect, and no single architecture's gain survives Holm correction. An uncorrected paired t -test would have given $p = 6 \times 10^{-5}$, which shows how much the naive test understates the variance.

The gain is not caused by test categories missing from training: every test category appeared in training in all 20 session-disjoint splits. The category-stratified protocol guarantees this by design and gains 0.14 to 0.19, which we do not count as extra evidence because it pushes rare-category sessions into a harder test set. A model-free probe points to a mechanism. Under a random split, 68% of the test windows have their nearest training window in the same session (65% to 71% across splits), although windows share no records; a neighbor drawn at random from same-category training windows would share the session 27% of the time, and one from all training windows 5%. The 1-NN lookup reaches a macro- F_1 of 0.958, against 0.814 under a session-disjoint split (Table 3). As every radio label is a session label, knowing the session gives the label, and a random split lets any model use this. The probe shows that session recognition is available and sufficient, not that the trained models rely on it.

Under the session-disjoint split, the lookup beats LR and DT and comes within 0.04 of the best supervised architecture, so on this layer the learned models add little to retrieving similar windows. The two sequence models reach 0.923 to 0.944 under a random split and 0.755 to 0.759 under a session-disjoint split, below the 1-NN lookup; their gain of 0.16 to 0.19 has corrected intervals that exclude zero for both (Holm-adjusted $p \leq 0.02$), helped by the smaller Holm correction of a family of two.

B. TIME ORDER AND BENIGN SESSIONS

The radio layer spans 52.96 days, but its sessions were captured in blocks (Fig. 3): nine benign sessions on days 0 to 6, the five attack categories on days 47 to 53, and a last benign session on day 53. Split by order, whole attack categories end up held out: for training fractions of 0.4 to 0.7, the latest sessions contain one to four attack categories never seen in training, plus one benign session. Such a split measures the detection of unseen categories, not temporal drift, and we do not report it as a drift estimate.

TABLE 3. Protocol sensitivity on the $\mathcal{D}_A$ radio layer: macro- F_1 under random, session-disjoint (SD) and category-stratified session-disjoint (CS) splits, and paired gains of the random split with Nadeau-Bengio corrected 95% intervals over 20 split seeds. No supervised architecture's gain is significant after Holm correction. 1-NN: a model-free nearest-neighbor lookup on the same splits.

Model	Random	Sess.-disj.	Cat.-strat.	Δ_{R-SD} [95% CI]	Δ_{R-CS} [95% CI]
LR	0.836	0.744	0.697	+0.093 [-0.05, +0.24]	+0.139 [-0.06, +0.33]
DT	0.950	0.782	0.758	+0.168 [+0.02, +0.32]	+0.192 [+0.02, +0.36]
RF	0.970	0.853	0.816	+0.117 [-0.04, +0.27]	+0.154 [-0.02, +0.32]
XGB	0.977	0.837	0.808	+0.140 [-0.00, +0.28]	+0.169 [-0.02, +0.35]
HGB	0.982	0.844	0.831	+0.138 [-0.01, +0.28]	+0.152 [-0.03, +0.33]
MLP	0.954	0.828	0.809	+0.126 [-0.01, +0.26]	+0.146 [-0.02, +0.31]
Strat.	0.498	0.507	0.500	-0.009 [-0.01, -0.00]	-0.002 [-0.01, +0.01]
Maj.	0.431	0.434	0.447	-0.002 [-0.02, +0.01]	-0.016 [-0.04, +0.01]
1-NN	0.958	0.814	0.794	+0.144 [+0.00, +0.28]	+0.164 [+0.01, +0.31]

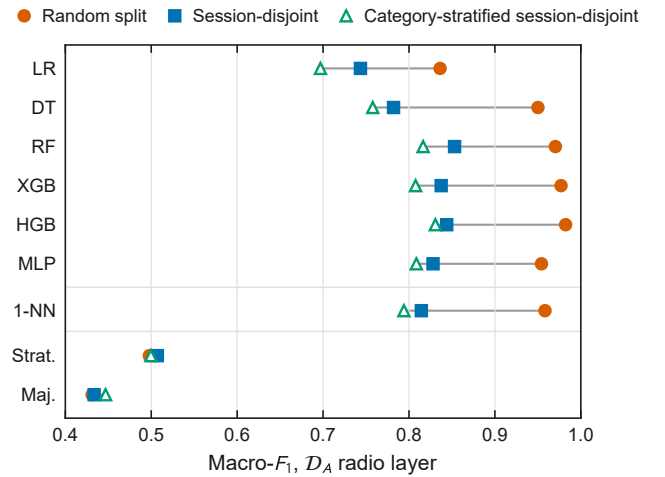


FIGURE 2. Macro- F_1 on the $\mathcal{D}_A$ radio layer under three protocols, mean over 20 split seeds; the gray line spans each model's three protocols. Every supervised model and the 1-NN lookup score higher under the random split, the input-blind baselines do not move, and keeping every attack category in training does not remove the gain. Intervals: Table 3.

To hold category coverage fixed, we held out the latest session of every category, benign included, and compared the result with 50 control draws that each hold out a random session per category. The latest sessions give a balanced accuracy of 0.52 to 0.57, close to chance, against control means of 0.75 to 0.83, yet 12% to 32% of the control draws score as low. The earliest give 0.95 to 0.99, which 4% to 34% of the control draws match or beat. With so few sessions, the design cannot separate an effect of time order from ordinary variation between sessions.

Much of that variation depends on which benign session is held out (Fig. 3). Averaged over the six architectures, six of the ten benign sessions give a false positive rate of at most 0.07, session 5 gives 0.14, and sessions 4 and 8 (days 3 and 6) give 0.81 and 0.98, so one first-week session is flagged more often than the last benign session, the only one from the attack period (0.87, or 196 false alerts per benign UE-hour). With ten benign sessions, a change over time cannot be told apart from this heterogeneity, and a detector's false positive rate on an unseen benign session of this testbed cannot be predicted

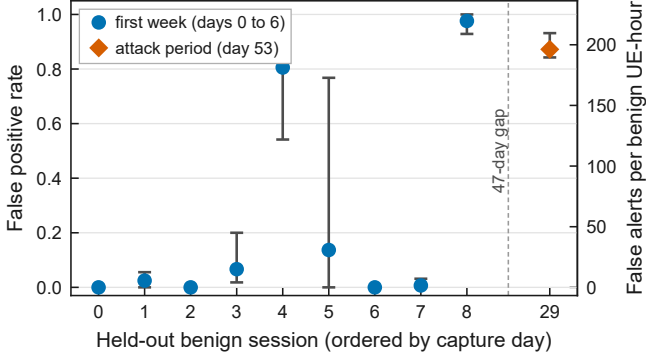


FIGURE 3. False positive rate on each benign radio session when that session alone is held out and the models train on the other 29: mean (marker) and range (bar) over the six architectures, sessions in capture order. The right axis gives the rate as false alerts per benign UE-hour (225 windows). Within one week, sessions range from no false positives to almost every window flagged, and one first-week session is flagged more often than the session recorded during the attack period.

from the sessions it was tested on.

C. THE TARGET CORPUS: LABELS AND A PUBLISHED RESULT

Label conflicts. For half of $\mathcal{D}_B$ the labels do not follow from the flow records: of the 1,215,889 flows, 52% (632,378) match a flow with the opposite label in every field except the row index, *Seq*, and *Offset* (Table 4, Fig. 4). All sit in the two UDP-flood files, 75% of file 5 (station 1) and 98% of file 15 (station 2). Each of the 281,529 flows labeled UDP flood in file 15 has a copy labeled benign in file 5 that agrees in every content field, and 33,704 of the 33,708 distinct records involved occur equally often in both files. The other 69,324 conflicting flows are floods of station 1's own attacker, labeled attack in file 5, with the same content as benign-labeled copies. The copies make up 59% of the benign class.

Which label is right? The full-field release carries addresses and lines up with the combined file row for row. All 281,525 benign copies are UDP flows from 10.155.15.7 to 10.41.150.68, the host pair, protocol, and time window of the flood labeled attack in file 15, whereas the attack labeled in file 5 comes from another host, 10.155.15.4, aimed at the same address. Station 1's capture thus appears to have recorded station 2's flood too, labeling only its own attacker; how is not documented, so this reading is our inference from the addresses. We therefore treat the copies as attack traffic with a wrong label and, rather than change published labels, report operating points on $\mathcal{D}_B$ (Table 10) and the transfer results of Section VI-D with and without them; Tables 8 and 11 and Fig. 8 use all flows.

These conflicts cap any classifier that reads only record content: even on the flows it was built from, the best function of the 90 content fields reaches a balanced accuracy of 0.766, and of the 18 shared columns 0.753, where DT, XGB, HGB, and the MLP land on a random split of $\mathcal{D}_B$ (Table 5). With *Seq* and *Offset*, almost every record becomes unique and the ceiling rises to 1.000.

TABLE 4. Label conflicts in $\mathcal{D}_B$ (5G-NIDD). Content: all fields of the published encoding except row index, *Seq*, *Offset*, and the labels. Ceiling: balanced accuracy of the best lookup on distinct records, in sample.

Features	Cols	Records	% in conflicting records			BA ceiling
			All	Benign	Attack	
Content fields	90	374 946	52.0	58.9	47.5	0.766
Shared space	18	246 281	62.4	59.7	64.1	0.753
Content + <i>Seq</i> , <i>Offset</i>	92	1 215 869	0.0	0.0	0.0	1.000

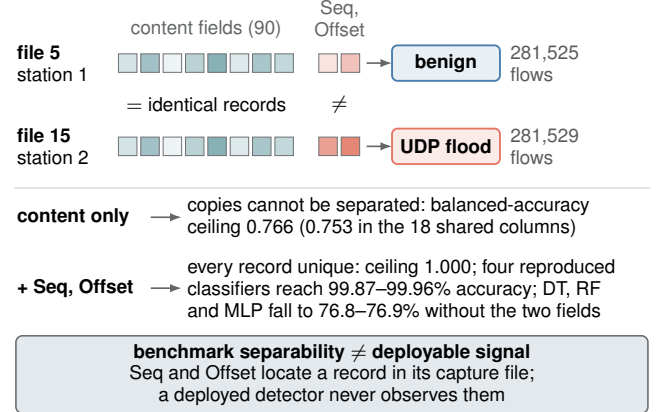


FIGURE 4. The label conflict in $\mathcal{D}_B$ (schematic tiles). Identical flow records occur in two capture files, one per base station, labeled benign in one and UDP flood in the other, so no function of the record content can separate them. They differ only in the row index and the record-position fields *Seq* and *Offset*, which make almost every record unique and account for the published 99.9% but are never seen by a deployed detector.

References within the target. To separate the cost of transfer from the limits of $\mathcal{D}_B$ itself, we trained and tested the six architectures on $\mathcal{D}_B$ alone (Table 5; one to three training samples per protocol, no intervals). A random split gives 0.708 to 0.752 in the shared space, the upper end being the ceiling above, and 0.842 to 0.925 once the benign copies leave the test set. Held-out capture files give 0.931 to 0.982 in the shared space and 0.990 to 1.000 with native features, though no held-out set contained a UDP-flood file. Across base stations, balanced accuracy drops to 0.618 to 0.641 from station 1 to station 2 and to 0.639 to 0.650 the other way. In the second direction a model flags the benign-labeled copies of station 1 and scores 0.973 to 0.988 once they leave the test set; in the first, the copies sit in the training data, so that a model trained with them labeled benign misses the floods of station 2 is our inference, not a tested result. Where labels follow from record content, the shared space separates the classes of $\mathcal{D}_B$ almost perfectly, so the shared features are not what limits transfer to $\mathcal{D}_B$.

A published result under the same ladder. The dataset authors report binary accuracies of 99.85% to 99.95% on a random 70/30 split [11]. We rebuilt their pipeline (the published one-hot encoding, the ten features with the highest ANOVA *F*-scores, *z*-score normalization, and their five classifiers with scikit-learn defaults, as tuned values are not reported). Our *F*-scores rank the features in their order, and under their protocol

TABLE 5. References inside $\mathcal{D}_B$: balanced accuracy of models trained and tested on $\mathcal{D}_B$ only, mean over n splits, each with a 240,000-flow training sample. *Range over the six architectures on the test flows without the benign copies of Table 4. Capture-file splits hold out four of the 20 files.

Features	Split	LR	DT	RF	XGB	HGB	MLP	BA*	n
Shared (18)	Random	0.708	0.752	0.723	0.752	0.752	0.750	0.84–0.92	3
Shared (18)	Capture file	0.955	0.982	0.940	0.943	0.956	0.931	0.93–0.98	3
Shared (18)	Station 1 to 2	0.618	0.638	0.640	0.641	0.640	0.635	0.62–0.64	1
Shared (18)	Station 2 to 1	0.639	0.648	0.648	0.648	0.648	0.650	0.97–0.99	1
Native	Random	0.710	0.764	0.735	0.764	0.764	0.764	0.80–0.87	2
Native	Capture file	0.990	1.000	0.999	1.000	1.000	0.994	0.99–1.00	2
Native	Station 1 to 2	0.698	0.661	0.662	0.662	0.662	0.662	0.66–0.70	1
Native	Station 2 to 1	0.648	0.648	0.649	0.650	0.650	0.654	0.99–1.00	1

we obtain 99.87% to 99.96% for DT, RF, KNN, and the MLP (Naive Bayes 92.71%), in line with their results (Table 6). Their two top features are the record-position fields `Seq` and `Offset`, which no deployed detector can observe and which, by the bounds above, are exactly what separates the conflicting records.

Swapping only these two fields for the next two in their ranking brings the accuracy of DT, RF, and the MLP down to 76.8% to 76.9% (balanced accuracy 0.704 to 0.706, false positive rate 0.588 to 0.591), yet RF still reaches 0.9999 on the test flows without the benign copies: the record-position fields let the published models tell identical records apart by their position in the file. With capture files held out, the published features give average balanced accuracies of 0.859 to 0.923 for DT, RF, and the MLP (RF 0.82 to 1.00 across splits), and without the record-position fields 0.799 to 0.800 (RF 0.50 to 1.00). Accuracy falls to 1.6% on the split that trains on one UDP-flood file and tests on the other, consistent with a model that learned the flood content as benign from the copies and so labels the held-out flood file, 98% flood, as benign. Across base stations the published features give 0.762 to 0.895, and those without the record-position fields 0.656 to 0.692. A 99.9% accuracy on this corpus says little about a new capture file or base station, and about a deployment, where record positions do not exist, it says nothing.

D. TRANSFER ACROSS CORPORA

Table 7 and Fig. 5 give the transfer from $\mathcal{D}_A$ to $\mathcal{D}_B$. In macro- F_1 every supervised model loses 0.10 to 0.13, but so does the feature-blind majority baseline (0.11), simply because $\mathcal{D}_B$ has a lower attack prevalence; net of it, the loss is -0.018 to +0.015, none significant after correction. Macro- F_1 is the wrong yardstick here, and the low source macro- F_1 values also reflect the 94.63% attack prevalence of $\mathcal{D}_A$ rather than weak detectors. Balanced accuracy, which prevalence leaves untouched, falls from 0.81–0.91 on held-out source data to 0.53–0.63 on $\mathcal{D}_B$, a drop of 0.21 to 0.36 that is significant for five of the six architectures after correction, positive in at least 18 of the 20 splits for every architecture, and 0.28 [0.15, 0.41] on average. The trivial baselines do not move, and target ROC-AUC lies between 0.53 and 0.70.

The detectors stay above chance on the target, but not by much: average target balanced accuracy is 0.57, and the corrected interval excludes 0.5 for four of the six architectures

(three after Holm correction). In macro- F_1 the stratified sampler scores 0.424 and a fair coin 0.494, a difference due only to the prior. LR transfers best on both prevalence-independent measures (balanced accuracy 0.63, ROC-AUC 0.70), while the MLP, strongest on held-out source data (0.91), reaches 0.55, among the lowest. With six architectures, the rank correlation between source and target balanced accuracy (-0.26, $p = 0.62$) is too imprecise to tell whether source performance predicts target performance, and the MLP shows that the best source model can be among the worst after transfer. The macro- F_1 rank correlation (0.83, $p = 0.04$) mixes discrimination with each model’s operating point under each corpus’s class balance and is equally imprecise. Our results agree in direction with Abraheem and Edhirig [18] (0.50 to 0.56 per attack family from NetsLab-5GORAN-IDD to 5G-NIDD); ours are pooled over attack types, so the sizes are not directly comparable.

Effect of the target labels. Some of this loss comes from the label conflict. A detector that learned to recognize a UDP flood on $\mathcal{D}_A$ finds the floods in file 15 and, by the same rule, flags the benign-labeled copies in file 5 (the five nonlinear architectures flag 0.70 to 0.85 of them, LR none). Without the copies (first 10 splits), target balanced accuracy is 0.61 to 0.78 (0.74 to 0.78 nonlinear, 0.61 for LR), and the loss against held-out source data shrinks to 0.00 to 0.25, significant for one of the six after correction. The conflicts reverse the ranking: LR looks best on all flows and is worst on the flows with consistent labels. Even there, transfer falls 0.06 to 0.25 short of the in-target reference on a random split (Table 5), and the stricter capture-file reference is higher still (0.931 to 0.982).

At $\tau = 0.5$, no architecture both keeps the benign traffic of $\mathcal{D}_B$ clear and catches most of its attacks. LR keeps a benign specificity of 0.98 but catches only 0.25 of the DoS traffic, while the tree models and the MLP catch more attacks and flag 0.43 to 0.56 of the benign-labeled traffic, which includes the flood copies; without them, their false positive rate is 0.03 to 0.18.

The reverse direction looks different (per-model values are in the repository). Balanced accuracy on $\mathcal{D}_A$ reaches 0.61 to 0.76, and the loss against the source reference (source minus target, optimistic because it comes from a random split) ranges from -0.037 to +0.145: LR gains 0.037, XGB, HGB, and the MLP change little, and DT and RF lose 0.09

TABLE 6. The 5G-NIDD authors’ pipeline [11] reproduced and moved one step per rung (R0: their protocol). Balanced accuracy on all $\mathcal{D}_B$ flows, and for RF also balanced accuracy and false positive rate without the benign copies. Pass: classifiers passing the deployability test on all flows / without the copies (Section VII-B). KNN runs at R0 only, as its prediction on later rungs dominated the runtime.

Rung	Split	Features	DT	Balanced accuracy, all flows				RF, w/o copies		Pass
				RF	KNN	NB	MLP	BA	FPR	
R0	Random 70/30	Published 10	0.9995	0.9996	0.9988	0.9310	0.9986	0.9997	0.0003	2 / 2
R1	Random 70/30	No Seq/Offset	0.7059	0.7059	–	0.6814	0.7042	0.9999	0.0003	0 / 2
R2	Capture file	Published 10	0.8594	0.9226	–	0.7501	0.8669	0.9578	0.0018	0 / 0
R2	Capture file	No Seq/Offset	0.7988	0.7996	–	0.7593	0.8004	0.8958	0.0013	0 / 0
R3	Base station	Published 10	0.7619	0.8949	–	0.6598	0.8577	0.9354	0.0009	0 / 1
R3	Base station	No Seq/Offset	0.6557	0.6555	–	0.6271	0.6919	0.8284	0.0007	0 / 0

TABLE 7. Transfer from $\mathcal{D}_A$ to $\mathcal{D}_B$ in the 18-column shared space, mean over 20 split seeds. Δ_{BA} : source minus target balanced accuracy, with Nadeau–Bengio corrected 95% interval; *: significant after Holm correction. Last column: macro- F_1 loss minus the majority baseline’s loss. Italic rows are the trivial baselines.

Model	Macro- F_1		Src.	Balanced accuracy		Target ROC-AUC	Δ_{F_1} vs. Maj.
	Src.	Tgt.		Tgt.	Δ_{BA} [95% CI]		
LR	0.626	0.530	0.837	0.627	+0.210* [+0.10, +0.32]	0.701	-0.018
DT	0.619	0.518	0.822	0.533	+0.290* [+0.08, +0.50]	0.526	-0.013
RF	0.701	0.572	0.808	0.581	+0.227 [-0.01, +0.47]	0.578	+0.015
XGB	0.646	0.532	0.864	0.562	+0.303* [+0.14, +0.47]	0.572	-0.001
HGB	0.647	0.537	0.867	0.566	+0.301* [+0.13, +0.47]	0.581	-0.004
MLP	0.655	0.531	0.912	0.553	+0.359* [+0.22, +0.50]	0.557	+0.010
<i>Strat.</i>	0.494	0.424	0.502	0.500	+0.001 [-0.01, +0.01]	0.500	-0.044
<i>Maj.</i>	0.492	0.378	0.500	0.500	+0.000 [+0.00, +0.00]	0.500	+0.000

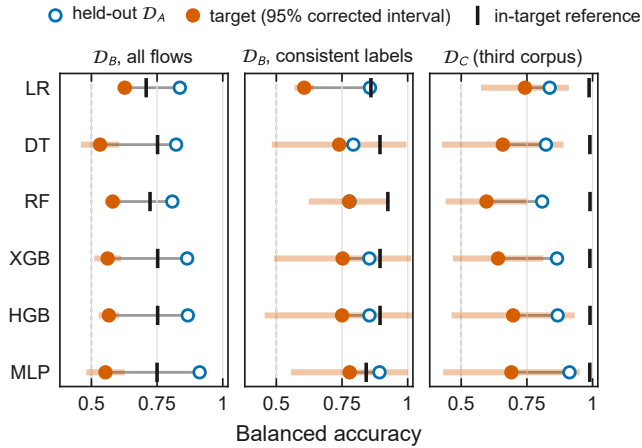


FIGURE 5. Balanced accuracy of detectors trained on $\mathcal{D}_A$, on held-out $\mathcal{D}_A$ (open) and on the target (filled, with corrected 95% interval): $\mathcal{D}_B$ on all flows (20 splits), $\mathcal{D}_B$ without the benign copies (10 splits), and $\mathcal{D}_C$ (20 splits). The vertical bar is the most optimistic in-target reference, a random split inside the target; transfer falls short of it everywhere, but on $\mathcal{D}_C$ comes close to a stricter reference (Section VI-F). The input-blind baselines sit on the dashed chance line.

to 0.14, the changes of LR, DT, and RF being significant after Holm correction. Macro- F_1 still falls sharply, because false negatives swamp the small benign class of $\mathcal{D}_A$ (94.63% attacks). Why a model trained on the narrower $\mathcal{D}_B$ separates $\mathcal{D}_A$ better than the reverse we have not tested; one possibility is that the larger benign class of $\mathcal{D}_B$, although 59% of it are flood copies, teaches a better model of benign behavior. Categories that $\mathcal{D}_B$ lacks are detected unevenly: the ensembles and the MLP recall at most 0.01 of the web attacks but 0.34

to 0.57 of the brute-force attacks.

E. COMPOSITION OF THE TRANSFER GAP

Separability. The domain classifier (Section IV-C) tells $\mathcal{D}_A$ from $\mathcal{D}_B$ flows with a balanced accuracy of 0.999 on the 18 shared columns and 0.997 on the nine exporter-robust ones. The largest shifts are in the protocol mix (most $\mathcal{D}_A$ flows are TCP, most $\mathcal{D}_B$ flows UDP), the direction ratios, and the packet counts (Fig. 6). The robust columns shift as much as the others ($\bar{W}_1$ of 0.28 against 0.27), pointing to a difference in traffic composition that no exporter would remove; whether flow segmentation also differs, the next test addresses.

A common exporter. The raw captures of $\mathcal{D}_B$ are public, so we re-extracted them with Zeek (version 8.0.10), the exporter family of $\mathcal{D}_A$, and mapped the records with the function used for $\mathcal{D}_A$. In every capture file the published attack flows belong to one host pair, and labeling a flow as attack when its host pair is an attack pair of the same file reproduces 100% of the published labels on the Argus flows, so we carried these labels over to the 346,882 Zeek flows (76% attacks). Zeek, which segments traffic differently (Section IV-B), produces about a third as many flows, with a higher attack share; we did not analyze which flows account for the difference. The flood copies reappear in the Zeek output (66,033 flows), so they are in the captures, not an artifact of the Argus export. We reran the transfer on the 10 splits of the Argus results without the copies, with identical trained models, so that only the export of the target changes.

The gap does not close. On all flows, balanced accuracy falls from 0.54–0.62 with Argus to 0.43–0.56 with Zeek, and the paired difference over splits (-0.15 to -0.06) has a

TABLE 8. Transfer from $\mathcal{D}_A$ to $\mathcal{D}_B$ under alternative feature sets, adaptation, and model families, mean over 20 split seeds. CORAL reads unlabeled target features.

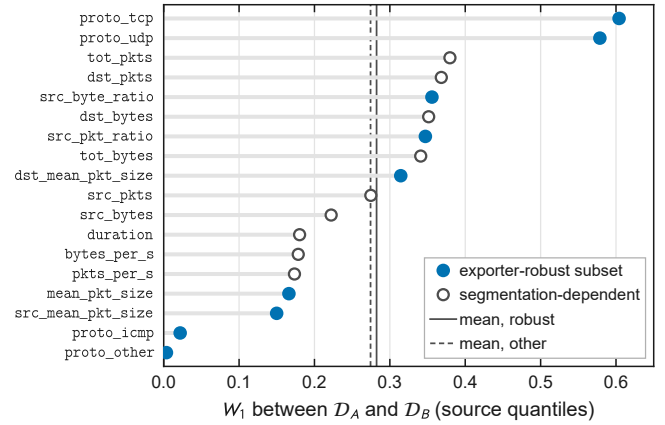
Arm	Model	Src. BA	Tgt. BA	Tgt. AUC	Tgt. FPR
Shared space (18)	LR	0.837	0.627	0.701	0.023
	DT	0.822	0.533	0.526	0.558
	RF	0.808	0.581	0.578	0.541
	XGB	0.864	0.562	0.572	0.429
	HGB	0.867	0.566	0.581	0.463
	MLP	0.912	0.553	0.557	0.539
Exporter-robust subset (9)	LR	0.804	0.510	0.583	0.854
	DT	0.883	0.480	0.488	0.177
	RF	0.894	0.507	0.469	0.206
	XGB	0.925	0.511	0.533	0.089
	HGB	0.939	0.521	0.541	0.090
	MLP	0.928	0.534	0.499	0.039
CORAL (reads target features)	LR	0.836	0.503	0.646	0.902
	DT	0.812	0.500	0.504	0.870
	RF	0.795	0.481	0.515	0.992
	XGB	0.906	0.473	0.478	0.906
	HGB	0.875	0.481	0.484	0.915
	MLP	0.930	0.602	0.632	0.552
Benign-only novelty	IF	0.503	0.492	0.464	0.024
	AE	0.455	0.479	0.484	0.074

95% paired t -interval below zero for all six architectures, uncorrected for split overlap since both arms use the same trained models. Without the copies, five architectures move by -0.06 to 0.04, none significantly, and LR loses 0.20, while the false positive rate rises for every architecture (0.13–0.46 against 0.03–0.18 with Argus). Two explanations remain, the compositional difference and differences between our Zeek configuration and the undocumented one behind $\mathcal{D}_A$, such as timeouts or single-packet handling (Section IX). The Zeek result cannot separate them; the robust-column shift favors composition without ruling out the second.

Model families and adaptation. None of three modifications run on the same splits produces a detector that transfers (Table 8). The nine exporter-robust columns alone lift held-out source balanced accuracy to 0.80 to 0.94 but leave the target at 0.48 to 0.53, no better than chance, which fits the compositional shift in these columns. CORAL, which reads unlabeled target features, gives 0.47 to 0.60; it helps only the MLP (0.60 against 0.55) and raises the target false positive rate to 0.55 to 0.99. The novelty detectors, trained on benign traffic alone, already fail on held-out source data, where the autoencoder falls below chance, likely (we did not test this) because floods and scans look like short benign flows in this feature space.

F. A THIRD CORPUS

To see whether these results hold beyond one pair of corpora, we added $\mathcal{D}_C$ as a second target, which changes the site, the flow exporter, and the vantage point together. Its labels are nearly consistent: only 3.2% of its flows sit on records that also occur with the opposite label, and the ceiling on balanced accuracy in the shared columns is 0.992. Trained and tested on $\mathcal{D}_C$, the six architectures reach 0.987 to 0.990 on a random split, and 0.660 to 0.826 when the capture file of one attack

**FIGURE 6.** Covariate shift between $\mathcal{D}_A$ and $\mathcal{D}_B$ per shared column, Eq. (2). Filled markers: the exporter-robust subset, fixed before any result under it was computed; vertical lines: the two group means. The robust columns shift as much as the rest, so the shift is not only a matter of how the exporters segment flows.

type is held out so that this type is never seen in training.

For transfer from $\mathcal{D}_A$ (Table 9, 20 splits), balanced accuracy on $\mathcal{D}_C$ ranges from 0.60 to 0.74 (ROC-AUC 0.70 to 0.93), close to the in-target reference that holds out an attack type (0.660 to 0.826), the fairer comparison because transfer, too, faces an unseen attack type, the PFCP attack. Every architecture detects the SYN flood, which training contained; recall is 0.10 to 1.00 on the ICMP flood and 0.00 to 0.99 on the PFCP attack. Most errors come from the benign traffic of the new site: the false positive rate is 0.225 to 0.807, and up to 0.81 of the core network's own signaling is flagged. The loss against held-out source data (0.09 to 0.22) varies so much between splits that none of it is significant after correction. From $\mathcal{D}_B$ (10 splits), balanced accuracy spans 0.42 to 0.85, ROC-AUC falls as low as 0.38 (a ranking inverted on $\mathcal{D}_C$), and the loss is significant for three of the six. At $\pi = 0.002$, the best operational precision at recall above 10% is 0.092 from $\mathcal{D}_A$ and 0.031 from $\mathcal{D}_B$, and no architecture passes the deployability test in either direction. Three findings thus repeat at a third site with a third exporter and nearly consistent labels: the SYN flood is caught, much benign traffic is flagged, and operational precision stays below 0.1.

G. ALERT BURDEN

Table 10 gives pooled operating points at $\tau = 0.5$ for four evaluation sets. On the radio layer, the six detectors have a pooled false positive rate of 0.22 to 0.33 (50 to 74 false alerts per benign UE-hour) and, at $\pi = 0.002$, an operational precision of 0.0055 to 0.0080, against a corpus precision of 0.90 to 0.93. These estimates are imprecise, because the false positive rate depends on which benign sessions enter the test set (Fig. 3): the bootstrap interval of the pooled rate spans 0.01 to 0.65 across architectures (3 to 146 false alerts per benign UE-hour), and that of operational precision 0.0028 to 0.13. Thirty sessions cannot pin down the radio alert burden; they show only that it depends on the benign conditions a detector

TABLE 9. Transfer to the third corpus $\mathcal{D}_C$ (5G core testbed, NFStream flows [55]): balanced accuracy on held-out source data and on $\mathcal{D}_C$, target ROC-AUC, recall per attack type, and the share of the core's background signaling flagged (Bkg.). Mean over 20 split seeds from $\mathcal{D}_A$ and 10 from $\mathcal{D}_B$, at $\tau = 0.5$.

Model	Src. BA	Tgt. BA	Trained on $\mathcal{D}_A$					Trained on $\mathcal{D}_B$	
			AUC	SYN	ICMP	PFCP	Bkg.	Tgt. BA	AUC
LR	0.837	0.742	0.800	1.00	0.10	0.00	0.27	0.517	0.726
DT	0.822	0.658	0.696	1.00	1.00	0.79	0.67	0.422	0.384
RF	0.808	0.596	0.926	1.00	1.00	0.99	0.81	0.540	0.697
XGB	0.864	0.640	0.888	1.00	1.00	0.74	0.70	0.688	0.814
HGB	0.867	0.698	0.932	1.00	1.00	0.80	0.61	0.565	0.632
MLP	0.912	0.691	0.837	1.00	0.20	0.61	0.44	0.848	0.766
Maj.	0.500	0.500	0.500	1.00	1.00	1.00	1.00	0.500	0.500

meets.

Flow-layer alert volumes are also large. The false positive rate is 0.066 to 0.201 on held-out $\mathcal{D}_A$ flows, or 15,726 to 48,289 false alerts per hour at $\lambda_b = 240,000$ benign flows per hour, and 0.023 to 0.558 on all flows of $\mathcal{D}_B$ (5,620 to 133,990 per hour). Without the benign copies it is 0.026 to 0.183, still 6,158 to 44,003 false alerts per hour, at an operational precision of 0.0072 to 0.0398.

On $\mathcal{D}_B$, raising the threshold barely improves operational precision (Fig. 7b). With the best threshold picked on the evaluation data, and recall above 10%, the upper bounds are 0.42 on held-out $\mathcal{D}_A$ flows and, on $\mathcal{D}_B$, 0.026 on all flows or 0.063 without the copies. On the radio layer a higher threshold does help (Fig. 7a), up to 0.036, but for four of the six architectures the bootstrap interval reaches 1.0: a resample that leaves out the benign sessions flagged almost entirely can leave thresholds with no false positive. The ratio of corpus to operational precision is arithmetic in π , from 2,627 at $\pi = 10^{-4}$ to 1.18 at 0.5, so an operational precision needs its assumed prevalence.

Estimation of operational precision. A split without a false positive yields $\text{PPV} = 1$ whatever its recall, so averaging PPV over splits rewards the architecture that draws the most such splits: on the radio layer the ratio between the best and worst architecture is 1.45 with pooled counts, 1.89 as a median over splits, and 16.0 as a mean. We use pooled counts throughout.

H. CALIBRATION AND PRIOR SHIFT

Under transfer the scores are poorly calibrated (Fig. 8). We fitted three calibrators on a group-disjoint source calibration fold (Table 11); whether one can move the reachable operating points depends on whether it keeps the order of the scores. Temperature scaling does, changing ROC-AUC by at most 2×10^{-5} . Platt scaling is monotone too, but its slope can be negative: for DT it inverted the ranking in one of 10 splits (ROC-AUC shift 0.95), because the tree's scores carried no ranking information on the calibration fold (ROC-AUC 0.48 on 96 held-out addresses) but ranked the test fold well (0.97), so the fitted slope of -0.55 inverted it; the small calibration fold is to blame, not Platt scaling. Isotonic regression, monotone but piecewise constant, merges scores into ties and yields the ROC convex hull on its fitting data [51]; on the target it lowered ROC-AUC by up to 0.21 and raised it by at most

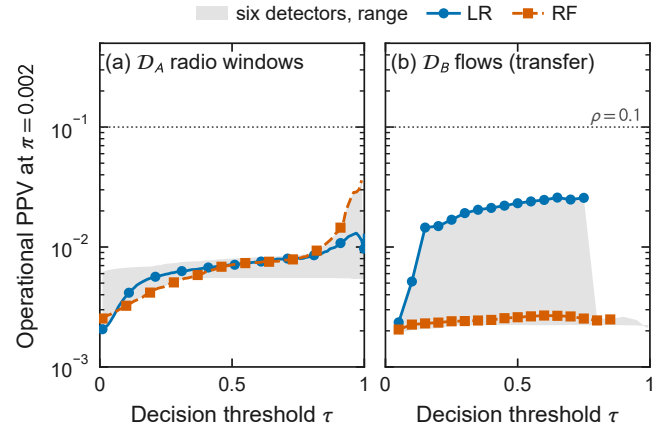


FIGURE 7. Pooled operational precision at $\pi = 0.002$ against the threshold on (a) the $\mathcal{D}_A$ radio layer and (b) all $\mathcal{D}_B$ flows after transfer, for recall ≥ 0.10 . Gray band: the six architectures. No threshold reaches the dotted precision bound $\rho = 0.1$ of the deployability test.

0.05. No calibrator raises target balanced accuracy for any architecture (changes of -0.12 to -0.01). Fitted on a source with 94.63% attacks, the calibrators pull the scores toward that prior, and the target false positive rate at $\tau = 0.5$ rises from 0.08–0.64 to 0.84–0.99 after Platt or isotonic calibration.

The usual remedy for prior shift is to rescale the scores to the target prior [47], which a deployment does not know. Expectation maximization [47] and black-box shift estimation [48] estimate it from unlabeled data, assuming that only the class balance changes, which fails here: for the published target prior of 0.607, the mean estimates per architecture run from 0.16 to 0.86, and 41% of the estimates fall below 0.05 or above 0.95. Even an oracle that rescales with the published prior does not beat the raw scores. The detectors are limited by discrimination on the target (ROC-AUC 0.53 to 0.70), not by calibration.

I. DECISION LATENCY

Table 12 gives per-call latency on one host for three implementations of each model: scikit-learn with a one-row pandas DataFrame, as our experiment code calls it; scikit-learn with a NumPy array on one thread; and ONNX Runtime on one thread, whose 12 exported models reproduce the decisions of scikit-learn exactly (decision agreement 1.0000). A call that does nothing has a 99th percentile of 0.001 ms, too little to

TABLE 10. Pooled operating points at $\tau = 0.5$ over 20 splits (“w/o copies”, $\mathcal{D}_B$ without the benign-labeled copies: first 10). FPR interval: 95% cluster bootstrap over splits and clusters (radio sessions, $\mathcal{D}_A$ source addresses, $\mathcal{D}_B$ capture files); for flows from the re-run on the first 10 splits. PPV at $\pi = 0.002$. [†]False alerts per benign UE-hour (radio, 225 windows per UE-hour) or per hour at $\lambda_b = 240,000$ benign flows (flows). [‡]Highest pooled PPV over all thresholds with recall at least 0.10.

Corpus	Model	Recall	FPR [95% CI]	Corpus prec.	PPV	False alerts [†]	Best PPV [‡]
$\mathcal{D}_A$ radio	LR	0.798	0.223 [0.017, 0.547]	0.921	0.0071	50	0.0130
	DT	0.910	0.329 [0.086, 0.639]	0.900	0.0055	74	0.0055
	RF	0.954	0.267 [0.024, 0.650]	0.921	0.0071	60	0.0360
	XGB	0.931	0.251 [0.013, 0.606]	0.924	0.0074	57	0.0116
	HGB	0.945	0.262 [0.018, 0.615]	0.922	0.0072	59	0.0086
	MLP	0.895	0.222 [0.019, 0.560]	0.929	0.0080	50	0.0109
	<i>Maj.</i>	1.000	1.000 [1.000, 1.000]	0.766	0.0020	225	0.0020
$\mathcal{D}_A$ flows	LR	0.892	0.201 [0.126, 0.445]	0.992	0.0088	48 289	0.0297
	DT	0.852	0.157 [0.068, 0.498]	0.994	0.0108	37 658	0.0120
	RF	0.920	0.198 [0.121, 0.611]	0.993	0.0092	47 568	0.4172
	XGB	0.884	0.090 [0.027, 0.373]	0.997	0.0193	21 625	0.2735
	HGB	0.881	0.066 [0.031, 0.112]	0.998	0.0262	15 726	0.2727
	MLP	0.905	0.071 [0.043, 0.349]	0.997	0.0250	16 996	0.2189
	<i>Maj.</i>	1.000	1.000 [1.000, 1.000]	0.967	0.0020	240 000	0.0020
$\mathcal{D}_B$ (transfer)	LR	0.277	0.023 [0.009, 0.079]	0.948	0.0232	5 620	0.0258
	DT	0.624	0.558 [0.111, 0.783]	0.633	0.0022	133 990	0.0022
	RF	0.703	0.541 [0.121, 0.796]	0.667	0.0026	129 843	0.0027
	XGB	0.552	0.429 [0.013, 0.687]	0.666	0.0026	102 890	0.0026
	HGB	0.596	0.463 [0.031, 0.699]	0.665	0.0026	111 104	0.0026
	MLP	0.646	0.539 [0.030, 0.745]	0.649	0.0024	129 320	0.0026
	<i>Maj.</i>	1.000	1.000 [1.000, 1.000]	0.607	0.0020	240 000	0.0020
$\mathcal{D}_B$ w/o copies	LR	0.269	0.057 [0.038, 0.086]	0.947	0.0094	13 629	0.0102
	DT	0.662	0.183 [0.103, 0.294]	0.931	0.0072	44 003	0.0077
	RF	0.719	0.165 [0.118, 0.258]	0.943	0.0087	39 561	0.0629
	XGB	0.531	0.026 [0.012, 0.047]	0.987	0.0398	6 158	0.0631
	HGB	0.548	0.047 [0.030, 0.080]	0.978	0.0227	11 315	0.0587
	MLP	0.612	0.053 [0.026, 0.102]	0.977	0.0225	12 794	0.0306
	<i>Maj.</i>	1.000	1.000 [1.000, 1.000]	0.790	0.0020	240 000	0.0020

TABLE 11. Calibration over 10 splits. Left: largest absolute change in ROC-AUC per calibrator, over splits and both evaluation sets. Right: target balanced accuracy at $\tau = 0.5$ for raw scores, Platt and isotonic calibration, and Platt followed by prior correction with the EM estimate or the published prior (oracle).

Model	max $ \Delta \text{ROC-AUC} $			Target balanced accuracy at $\tau = 0.5$				
	Temp.	Platt	Iso.	Raw	Platt	Iso.	+EM	+Oracle
LR	$< 10^{-6}$	1×10^{-6}	0.209	0.623	0.514	0.506	0.512	0.600
DT	2×10^{-5}	0.946	0.170	0.554	0.505	0.504	0.525	0.526
RF	$< 10^{-6}$	$< 10^{-6}$	0.050	0.590	0.564	0.548	0.531	0.561
XGB	$< 10^{-6}$	$< 10^{-6}$	0.150	0.577	0.535	0.494	0.533	0.562
HGB	$< 10^{-6}$	$< 10^{-6}$	0.095	0.571	0.517	0.500	0.543	0.553
MLP	2×10^{-5}	2×10^{-5}	0.045	0.583	0.499	0.499	0.540	0.564

distort the comparisons.

Implementation matters more than architecture (Fig. 9). For LR, the DataFrame path takes 0.44 ms at the median against 0.11 ms for an array, a difference we attribute to input validation without having measured it. RF takes 5.6 ms on one thread and 0.012 ms in ONNX Runtime, and 29 ms through its default thread pool, a configuration not in Table 12. On the radio layer the 99th percentile under ONNX Runtime lies between 0.014 and 0.035 ms for all architectures, while scikit-learn with arrays reaches 21.5 ms. Ranking architectures by latency mostly ranks implementations, so latencies from other hosts or languages cannot rank them. For context only, Obiuevwi et al. report microsecond inference for logistic regression and a small MLP inside an OpenAirInterface

and FlexRIC near-real-time RIC [19], within an order of magnitude of our ONNX Runtime figures.

Aggregating one 16-record window takes 0.064 ms at the 99th percentile, and the upper bound of the 99th percentile of aggregation plus ONNX inference is at most 0.12 ms for every architecture, within any budget in the specified range. A repeat that recorded every call for Fig. 9 gave medians of 1.02 to 2.1 times those in Table 12 (median ratio 1.46), as without CPU isolation latency depends on the state of the host, but that bound stayed at or below 0.36 ms.

Table 13 adds the remaining terms of Eq. (5) for window-level detectors in the FlexRIC loop of Section V. The six models ran with a report every 10 ms (5,000 reports each after 200 warm-up reports), LR and RF also every 1 ms (20,016

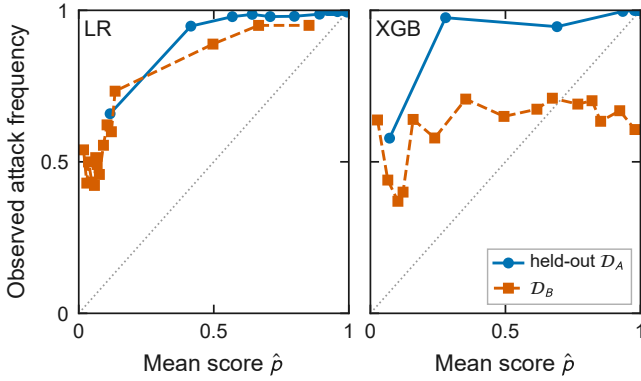


FIGURE 8. Reliability diagrams for LR and XGB on held-out $\mathcal{D}_A$ and all $\mathcal{D}_B$ flows, fifteen equal-mass bins pooled over the 20 transfer splits (Table 11 uses 10 splits of its own three-way split). Points above the diagonal mean that scores understate the attack frequency; on $\mathcal{D}_B$ the XGB curve stays near 0.4 to 0.7 at every score.

after 1,000 warm-up reports), with 7.7 UEs per report on average, all scored. At the 10 ms period, delivery from the E2 node to the xApp takes 1.64 ms at the median and the control round trip 0.78 ms. Inference in the loop needs at most 0.229 ms at the 99th percentile, several times the standalone figures; the loop uses the C API under WSL2 rather than the Python API on Windows, and we did not isolate which difference is responsible. Delivery and control account for at least 86% of the mean loop time for every model, so once a RIC is in the path the model barely matters for latency. The full loop, from report to acknowledged control message, takes 4.45 to 4.95 ms at the 99th percentile (upper 95% bound at most 5.01 ms) and at most 5.71 ms at the 99.9th, all within the smallest specified budget of 10 ms on this emulated loop. The largest single value at this period was 11.2 ms.

The loop ran faster at the 1 ms period: its 99th percentile fell to 1.42 to 1.69 ms, with at most 4 queued reports, no dropped report, and every term faster, inference included (at most 0.050 ms). We did not control CPU power management, and a host that idles between sparse reports and pays to wake up would produce this pattern.

For a flow-level detector the expensive part is feature construction, not inference. On the three DDoS captures of $\mathcal{D}_A$, our vectorized exporter (Section V) produces flow records at 0.20 to 0.90 ms per flow, 22.8 to 41.6 times faster than our per-packet reference implementation (4.49 to 37.34 ms per flow), at a cost that grows with the packets per flow. On the two smaller captures, where re-running the reference is affordable, both agree exactly in packets and bytes on all 38,032 shared IPv4 flow keys (the reference also emits 22 and 41 IPv6 link-local keys, which the IPv4-only vectorized exporter skips). Mapping a record onto the shared space with our pandas code adds 8.5 ms at the median, against at most 0.037 ms for ONNX inference at the 99th percentile. Export and mapping alone thus take close to 10 ms before any transport to the xApp, which our loop lacks, so we have no measured latency term for flow-level detectors and cannot say what a native user-plane exporter would cost.

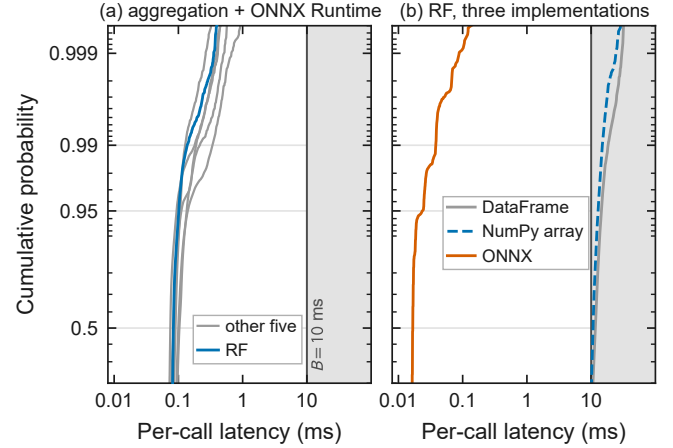


FIGURE 9. Empirical distribution of per-call radio-layer latency on one host, 20,000 calls per stage, on a logit scale. (a) Window aggregation plus ONNX Runtime inference, six architectures. (b) RF under the three implementations of Table 12. Shaded: beyond $B = 10$ ms. Only feature construction and inference are timed (Table 13: full RIC loop). The curves come from a repeat that kept every call; Table 12 reports the original run.

VII. DISCUSSION

A. IMPLICATIONS FOR O-RAN PRACTICE

Data path. E2SM-KPM delivers aggregated measurements once per reporting period, not packets, so a window-level detector can live inside the RIC, whereas a flow-level detector needs an exporter near the user plane (in the central unit or at the user-plane function) and a way to move its records to the xApp, such as a custom E2 service model or the non-real-time data collection path. That transport adds to t_{ind} in Eq. (5), which our RIC loop measures for KPM reports only.

Detection delay. Computation is a small part of detection time: a window-level decision waits for its 16 s window to end, a flow-level one for the flow record, which for a long connection means an idle or active timeout. Computational latency should be reported with the KPM reporting period and flow timeouts.

Acting on alerts. An E2SM-RC action such as isolating or steering a UE hits a falsely accused UE as hard as a guilty one. At the rates in Table 10, automatic mitigation on the radio layer would act against an average benign UE 50 to 74 times an hour, far more often in some benign sessions than in others (Fig. 3). Alerts of this quality can inform an operator, or a policy set through the A1 interface of the non-real-time RIC, but should not drive automatic control, and for such detectors the 10 ms budget matters less than detection quality: latency binds only for a detector whose alerts are good enough to act on automatically.

Admission and monitoring. The deployability test could serve as an admission check at a new site, given labeled traffic from it. Afterwards, unsupervised shift detection [52] can flag traffic that departs from the training distribution; as a supervised classifier separates our corpora with a balanced accuracy of 0.999, such a monitor would likely detect this shift, though we did not test one.

TABLE 12. Per-call latency in milliseconds on one host (Section V): 20,000 calls after 1,000 warm-up calls, one thread per library, garbage collection enabled. No RIC in the path (Table 13 gives the loop through a RIC). Radio path $\bar{p}_{99}$: upper 95% bound of the 99th percentile of window aggregation plus ONNX inference.

Model	sklearn, DataFrame		sklearn, array		ONNX Runtime		Radio path $\bar{p}_{99}$	Flow layer, p_{99} array ONNX	
	p_{50}	p_{99}	p_{50}	p_{99}	p_{50}	p_{99}			
LR	0.44	0.72	0.11	0.17	0.010	0.023	0.12	0.16	0.025
DT	0.31	0.52	0.03	0.04	0.010	0.014	0.11	0.06	0.024
RF	6.25	12.99	5.57	21.50	0.012	0.024	0.10	4.45	0.029
XGB	0.73	1.00	0.16	0.32	0.011	0.021	0.09	0.40	0.024
HGB	1.48	2.72	1.17	1.92	0.013	0.035	0.12	1.22	0.034
MLP	0.43	0.67	0.11	0.21	0.017	0.035	0.11	0.15	0.037

TABLE 13. Decision loop of a window-level detector through FlexRIC with an emulated E2 node on one host, in ms, ranges over models, at a 10 ms or 1 ms reporting period. Terms of Eq. (5): delivery to the xApp (t_{ind}), queueing (t_q), aggregation (t_{feat}) and inference (t_{inf}) for one UE, and control up to acknowledgment (t_{act}). "Loop": KPM header timestamp to acknowledgment, for the first UE of a report or all its UEs.

Term	10 ms, six models		1 ms, LR and RF	
	p_{50}	p_{99}	p_{50}	p_{99}
t_{ind}	1.51–1.86	3.02–3.05	0.51–0.53	0.84–0.98
t_q	0.063–0.071	0.18–0.21	0.015–0.016	0.069
t_{feat}	0.001	0.002–0.003	<0.001	0.001
t_{inf}	0.050–0.12	0.13–0.23	0.014–0.025	0.038–0.050
t_{act}	0.72–0.87	1.40–1.50	0.20–0.21	0.53–0.60
Loop, 1 UE	2.38–2.97	4.36–4.59	0.76–0.79	1.38–1.59
Loop, all	2.48–3.18	4.45–4.95	0.77–0.86	1.42–1.69

B. APPLYING THE DEPLOYABILITY TEST

No O-RAN specification defines the bounds of Eq. (6), so we chose them ourselves; an operator would set them from its own traffic and staffing. The generalization bound is $\delta = 0.10$ in balanced accuracy, applied to the upper end of the corrected 95% interval. The alert term asks for a threshold that detects at least half of the attacks ($r_{min} = 0.5$), at which at least one alert in ten is real ($\rho = 0.1$ at $\pi = 0.002$), and which raises at most $A_{max} = 200$ false alerts per hour at $\lambda_b = 240,000$ benign flows per hour. All alert conditions are checked on $\mathcal{D}_B$, where a detector would be deployed, with the threshold searched there too, so the alert term is the best a detector could reach. For latency we take $B = 10$ ms, the strictest budget the specification allows. It was measured through a RIC only for window-level detectors with the same architectures (99th-percentile upper bound at most 5.01 ms), not for the flow-level detectors tested here (Section VI-I), so Table 14 reports B_{min} for the window-level loop and the verdict rests on the other two terms.

Table 14 gives the verdict on all flows: no architecture passes (zero of six). All fail the generalization term, with upper bounds of 0.32 to 0.50, and all fail the alert term by a wide margin: at a recall of at least 0.5, the best operational precision on $\mathcal{D}_B$ is 0.0027, and the fewest false alerts per hour are 95,507. Each one of ρ above 0.0027, A_{max} below 95,507 per hour, or δ below 0.32 would on its own keep every architecture out. Nor does the verdict depend on the declared prevalence: the false-alert count does not involve π and exceeds A_{max} at every threshold, and the precision

TABLE 14. Deployability test of Eq. (6) on all $\mathcal{D}_B$ flows. $\bar{\Delta}_{BA}$: upper 95% bound of the balanced-accuracy loss (pass if ≤ 0.10). Best PPV and fewest false alerts per hour at recall ≥ 0.5 (pass if PPV ≥ 0.1 and alerts ≤ 200). $^\dagger B_{min}$: smallest of 1, 2, 5, 10 ms met by the upper bound of the 99th percentile of the FlexRIC loop, for window-level detectors of the same architecture; the flow-level latency term was not measured and does not enter $\mathcal{P}$.

Model	$\bar{\Delta}_{BA}$	Gen.	Best PPV	Min. alerts/h	Alert	$B_{min}^\dagger$ (ms)	$\mathcal{P}$
LR	0.32	×	0.0024	166 328	×	5	0
DT	0.50	×	0.0022	118 342	×	5	0
RF	0.47	×	0.0027	98 410	×	5	0
XGB	0.47	×	0.0026	95 507	×	5	0
HGB	0.47	×	0.0026	99 277	×	10	0
MLP	0.50	×	0.0026	105 744	×	5	0

condition alone would need $\pi \geq 0.076$, 38 times the declared value.

On the flows without the benign copies, with both terms computed on those flows, the verdict is also zero of six, but the generalization term fails for a different reason: the loss is significant for only one architecture (Section VI-D), and the upper bounds of 0.29 to 0.41 exceed δ because the intervals are wide, not because a loss of that size is established. The alert term decides the verdict. At a recall of at least 0.5 the best operational precision is 0.0419 and the fewest false alerts per hour 5,603; the precision condition alone would hold from $\pi = 0.005$, but the false-alert count rules out every architecture.

Fig. 10 shows how far the bounds would have to move. On the flows without the copies, an architecture meets the alert term only if an operator accepts several thousand false alerts per hour at a precision of a few percent, and the generalization term only if δ exceeds 0.29 (0.32 on all flows). Latency would matter only once the first two terms were met; in our window-level loop it clears the smallest specified budget by about a factor of two.

Attainability of the test. A test that nobody passes may just be too strict. The published 5G-NIDD pipeline (Section VI-C) shows where it can be passed, with two caveats: its generalization term is the drop from each classifier's own R0 score, a point estimate over two splits that is zero at R0 by construction, and its latency was not measured. On its own random split, two of its five classifiers pass (Table 6), with operational precision up to 0.79 and as few as 128 false alerts per hour, and on all flows none passes at a later rung. On the

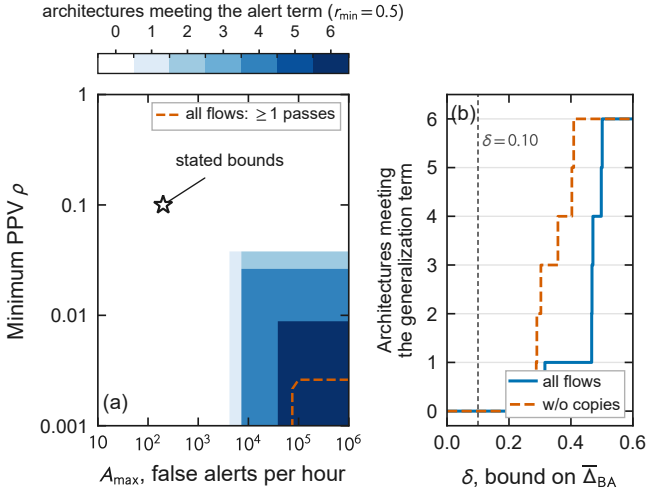


FIGURE 10. Sensitivity of the deployability test to its bounds. (a) Architectures meeting the alert term ($r_{\min} = 0.5$, $\pi = 0.002$, $\lambda_b = 240,000$ flows per hour) on $\mathcal{D}_B$ without the benign copies as $A_{\max}$ and ρ vary; on all flows, one or more pass only below and right of the dashed line. Star: our bounds. (b) Architectures meeting the generalization term as δ varies. Passing requires both terms.

flows without the copies, DT and RF still pass at R1 (random split, no record-position fields), and RF passes across base stations (R3) with the published features, which a deployment would not have. None passes with capture files held out (R2), nor at R3 without the record-position fields, where DT and RF meet the alert term but lose 0.17 in balanced accuracy. The test can thus be passed within one corpus once the label conflict is removed, but not when whole capture files are held out. Under the evaluations closest to deployment that the available corpora allow, transfer to an independently collected corpus, the detectors in this paper fail.

C. RELATION TO REPORTED RESULTS

The 5G-NIDD results we cite [11], [12] are in-distribution scores, and under the same split ours are as high: 99.87% to 99.96% accuracy for the reproduced pipeline and 0.987 to 0.990 balanced accuracy inside $\mathcal{D}_C$. Setting such numbers beside transfer results would compare protocols, not detectors. In our transfer tests, a held-out score under a random split did not predict performance on another corpus: on the radio layer it rose with session recognition, on 5G-NIDD it rested on record-position fields, and the best source model was among the worst after transfer. The reporting requirements of Section X list the quantities that changed our conclusions.

VIII. REPRODUCIBILITY AND ETHICS

Code, configuration, seeded splits, per-split results, and the generators for every table, figure, and in-text number are at <https://github.com/adeliusa486/oran-ids-deployability>, where one script regenerates and checks every number or re-runs each experiment from the SHA-256-verified raw corpora (REPRODUCE.md); it builds FlexRIC, the xApp, and Zeek without administrator rights on Linux or WSL2. $\mathcal{D}_A$ [36] and

$\mathcal{D}_B$ [34], including the full-field release and the GTP-removed captures, are distributed by their authors under CC-BY-4.0. $\mathcal{D}_C$ is available from its authors' repository [55], which states no license, so we do not redistribute it. All traffic comes from testbed devices and test networks, and no personal data was processed. Showing when detectors fail carries a modest dual-use risk, which we judge smaller than the benefit to operators of knowing that reported accuracy need not carry over to their networks.

IX. LIMITATIONS

Flow exporters. Zeek, Argus, and NFStream produced $\mathcal{D}_A$, $\mathcal{D}_B$, and $\mathcal{D}_C$. Re-extracting $\mathcal{D}_B$ with Zeek did not close the gap (Section VI-E), but the Zeek version and configuration behind $\mathcal{D}_A$ are undocumented, so our export may still differ from it in timeouts or in the handling of single-packet flows. The Zeek flows take their labels through the host pairs of each capture; the rule reproduces every published label on the Argus flows, but would give the attack label to a Zeek flow whose host pair carries both attack and benign traffic within one capture. $\mathcal{D}_C$ was not re-extracted.

Target labels. We treat the benign flows of $\mathcal{D}_B$ that are content copies of UDP-flood records (59% of the benign class) as mislabeled because their addresses point to the attacker at the other base station (Section VI-C), an inference from the addresses alone, so every operating point on $\mathcal{D}_B$ appears with and without them. The recovered capture files match the per-station files of the full-field release in row count and attacking host (all 10 files of station 1).

Radio capture schedule. Nine of the ten benign radio sessions were recorded six weeks before the first attack session, mostly with one UE, and every radio label is a session label, so a radio detector may learn capture periods as well as attacks. The one benign session from the attack period is flagged almost entirely, but so are two early sessions (Fig. 3), and ten benign sessions are too few to decide whether the detectors learned capture periods or the benign sessions simply differ from each other.

Number of corpora. Three corpora show that the transfer gap recurs, not how it varies across sites. The third is a 5G core testbed, not an O-RAN deployment, with only flood attacks, one control-plane attack, and three capture files, too few for cluster-based intervals. We found no other O-RAN corpus with paired radio telemetry.

Latency. All latency figures come from one workstation without CPU isolation, and the RIC loop used an emulated E2 node over loopback SCTP, with no radio and no network between sites. It measures the processing cost of the RIC software path, which also depends on power management, not the delay of a deployment, and covers window-level detectors only. CPU and memory under load were not measured.

Operational parameters and the deployability test. The prevalence $\pi = 0.002$ and the rate λ_b are declared, not measured on an operational network, and we found no measured source for π . Only π enters operational precision, and alert counts scale linearly with λ_b . The test's bounds are our own

choice; Section VII-B shows the verdict under other bounds and prevalences, but an operator may weigh recall, precision, and alert volume differently, and the test ignores detection delay and the cost of a missed attack.

Sample size. The radio layer has 30 sessions, each benign or from one attack category, so session and category are confounded, and the protocol and temporal analyses detect only large effects. With ten benign sessions, the radio false positive rate and operational precision carry wide intervals (Section VI-G). The transfer statistics vary the training set over 20 splits but test on one fixed target corpus, so they miss the variability across other target corpora.

Scope. We study binary detection only. Multi-class attribution, feature attribution [62], and the fusion of radio and flow features, which needs the raw archives [28] because the summary files cannot be joined, remain open. Three attack categories of $\mathcal{D}_A$ are missing from $\mathcal{D}_B$ and can be tested in one direction only.

X. CONCLUSION AND REPORTING REQUIREMENTS

A held-out score on one corpus says little about how a detection xApp will behave on data from another site. On the radio layer of NetsLab-5GORAN-IDD, a random split raised macro- F_1 by 0.13 on average (95% interval -0.01 to 0.27), and a model-free lookup gained about as much, consistent with the recognition of capture sessions. Half of 5G-NIDD lies on records that occur under both labels, the addresses of the benign copies point to the attacker at the other base station, and the published 99.9% depends on the record-position fields that tell the copies apart. On flows with consistent labels, transferred detectors reach a balanced accuracy of 0.61 to 0.78, below what the target corpus itself allows, although the loss against held-out source data is significant for only one architecture; on a third corpus they catch the SYN flood but flag much benign traffic, with no significant loss against source data. At the declared prevalence of 0.002, no transferred detector reaches an operational precision of 0.1 at any threshold with recall above 10%. Only window-level detectors meet the latency term, at 4.45 to 4.95 ms at the 99th percentile through a FlexRIC loop with an emulated E2 node on one host, a measurement rather than a conformance result that leaves out the detection delay of a 16 s window or a flow timeout. We therefore propose that any study claiming a deployable O-RAN detection capability report at least the following:

- 1) performance on an independently collected corpus in a prevalence-invariant metric (balanced accuracy or ROC-AUC), beside input-blind baselines and a group-disjoint reference trained inside that corpus;
- 2) the split protocol and group key, group-disjoint results with intervals that account for overlap between resampled splits, and the category coverage of any time-based split;
- 3) the benign false positive rate with its unit, the resulting alert rate, and the operational precision at a stated

base rate, from pooled confusion counts with cluster-resampling intervals;

- 4) the operating threshold, how it was chosen, and the recall it gives up;
- 5) tail latency from indication to control acknowledgment, with stages, implementation, reporting period, and host, measured through a RIC and, before any conformance claim, with a real E2 node;
- 6) evidence that the labels are a function of the features used, with the share of identical records under opposite labels, and that no record-position or identifier field is a feature.

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REFERENCES

- [1] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "Understanding O-RAN: Architecture, interfaces, algorithms, security, and research challenges," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 2, pp. 1376–1411, 2023. doi: 10.1109/COMST.2023.3239220
- [2] A. S. Abdalla, P. S. Upadhyaya, V. K. Shah, and V. Marojevic, "Toward next generation open radio access networks: What O-RAN can and cannot do!" *IEEE Network*, vol. 36, no. 6, pp. 206–213, 2022. doi: 10.1109/MNET.108.2100659
- [3] S. Niknam, A. Roy, H. S. Dhillon, S. Singh, R. Banerji, J. H. Reed, N. Saxena, and S. Yoon, "Intelligent O-RAN for beyond 5G and 6G wireless networks," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2022, pp. 215–220. doi: 10.1109/GCWkshps56602.2022.10008676
- [4] L. Bonati, S. D'Oro, M. Polese, S. Basagni, and T. Melodia, "Intelligence and learning in O-RAN for data-driven NextG cellular networks," *IEEE Communications Magazine*, vol. 59, no. 10, pp. 21–27, 2021. doi: 10.1109/MCOM.101.2001120
- [5] M. Antonakakis, T. April, M. Bailey, M. Bernhard, E. Bursztin, J. Cochran, Z. Durumeric, J. A. Halderman, L. Invernizzi, M. Kallitsis, D. Kumar, C. Lever, Z. Ma, J. Mason, D. Menscher, C. Seaman, N. Sullivan, K. Thomas, and Y. Zhou, "Understanding the Mirai botnet," in *Proc. 26th USENIX Security Symposium*, 2017, pp. 1093–1110. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/antonakakis>
- [6] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "DDoS in the IoT: Mirai and other botnets," *Computer*, vol. 50, no. 7, pp. 80–84, 2017. doi: 10.1109/MC.2017.201
- [7] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "CoO-RAN: Developing machine learning-based xApps for open RAN closed-loop control on programmable experimental platforms," *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 5787–5800, 2023. doi: 10.1109/TMC.2022.3188013
- [8] R. Schmidt, M. Irazabal, and N. Nikaein, "FlexRIC: An SDK for next-generation SD-RANs," in *Proc. ACM CoNEXT*, 2021, pp. 411–425. doi: 10.1145/3485983.3494870
- [9] H. Wen, P. Porras, V. Yegneswaran, A. Gehani, and Z. Lin, "5G-Spector: An O-RAN compliant layer-3 cellular attack detection service," in *Proc. Network and Distributed System Security Symposium (NDSS)*, 2024. doi: 10.14722/ndss.2024.24527
- [10] A. Scalingi, S. D'Oro, F. Restuccia, T. Melodia, and D. Giustiniano, "Det-RAN: Data-driven cross-layer real-time attack detection in 5G open RANs," in *Proc. IEEE Conf. on Computer Communications (INFOCOM)*, 2024, pp. 41–50. doi: 10.1109/INFOCOM52122.2024.10621223
- [11] S. Samarakoon, Y. Siriwardhana, P. Porambage, M. Liyanage, S.-Y. Chang, J. Kim, J. Kim, and M. Ylianttila, "5G-NIDD: A comprehensive network intrusion detection dataset generated over 5G wireless network,"

- arXiv:2212.01298, 2022. [Online]. Available: <https://arxiv.org/abs/2212.01298>
- [12] L. Ilias, G. Doukas, V. Lamprou, C. Ntanos, and D. Askounis, "Convolutional neural networks and mixture of experts for intrusion detection in 5G networks and beyond," arXiv:2412.03483v2, 2025. [Online]. Available: <https://arxiv.org/abs/2412.03483v2>
- [13] S. Layeghy and M. Portmann, "Explainable cross-domain evaluation of ML-based network intrusion detection systems," *Computers and Electrical Engineering*, vol. 108, p. 108692, 2023. doi: 10.1016/j.compeleceng.2023.108692
- [14] S. Layeghy, M. Gallagher, and M. Portmann, "Benchmarking the benchmark: Comparing synthetic and real-world network IDS datasets," *Journal of Information Security and Applications*, vol. 80, p. 103689, 2024. doi: 10.1016/j.jisa.2023.103689
- [15] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Transactions on Information and System Security*, vol. 3, no. 3, pp. 186–205, 2000. doi: 10.1145/357830.357849
- [16] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *Proc. IEEE Symposium on Security and Privacy*, 2010, pp. 305–316. doi: 10.1109/SP.2010.25
- [17] ETSI, "Publicly available specification (PAS); E2 interface: General aspects and principles (O-RAN.WG3.E2GAP-R003-v04.01)," European Telecommunications Standards Institute, ETSI TS 104 038 V4.1.0, Oct. 2024. [Online]. Available: https://www.etsi.org/deliver/etsi_ts/104000_104099/104038/04.01.00_60/ts_104038v040100p.pdf
- [18] A. Abrahim and A. Edhirig, "Bidirectional cross-dataset generalization and label-efficient adaptation for 5G network intrusion detection," *Wadi Alshatti University Journal of Pure and Applied Sciences*, vol. 4, no. 2, Aug. 2026. [Online]. Available: <https://waujpas.com/index.php/journal/article/view/509>
- [19] L. Obiuevwi, K. J. Rechowicz, S. Jayarathna, S. H. Bouk, F. Afrin, C. N. Barati, N. Moghim, V. Nannou, M. E. Ur Rahman, and S. Shetty, "Enabling real-time AI in O-RAN: Deploying and measuring AI inside a near-RT RIC xApp," in *Proc. 35th Int. Conf. on Computer Communications and Networks (ICCCN)*, Jul. 2026, pp. 1–9. doi: 10.1109/ICCCN.2026.11662852
- [20] L. Bonati, M. Polese, S. D'Oro, S. Basagni, and T. Melodia, "OpenRAN Gym: AI/ML development, data collection, and testing for O-RAN on PAWR platforms," *Computer Networks*, vol. 220, p. 109502, 2023. doi: 10.1016/j.comnet.2022.109502
- [21] L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "SCOPE: An open and softwareized prototyping platform for NextG systems," in *Proc. ACM MobiSys*, 2021, pp. 415–426. doi: 10.1145/3458864.3466863
- [22] N. Nikaein, M. K. Marina, S. Manickam, A. Dawson, R. Knopp, and C. Bonnet, "OpenAirInterface: A flexible platform for 5G research," *ACM SIGCOMM Computer Communication Review*, vol. 44, no. 5, pp. 33–38, 2014. doi: 10.1145/2677046.2677053
- [23] I. Gomez-Migueluez, A. Garcia-Saavedra, P. D. Sutton, P. Serrano, C. Cano, and D. J. Leith, "srsLTE: An open-source platform for LTE evolution and experimentation," in *Proc. ACM WiNTECH*, 2016, pp. 25–32. doi: 10.1145/2980159.2980163
- [24] ETSI, "Publicly available specification (PAS); O-RAN security threat modeling and risk assessment (O-RAN.WG11.Threat-Modeling.O-R003-v03.00)," European Telecommunications Standards Institute, ETSI TR 104 106 V3.0.0, Jun. 2025. [Online]. Available: https://www.etsi.org/deliver/etsi_tr/104100_104199/104106/03.00.00_60/tr_104106v030000p.pdf
- [25] M. Liyanage, A. Braeken, S. Shahabuddin, and P. Ranaweera, "Open RAN security: Challenges and opportunities," *Journal of Network and Computer Applications*, vol. 214, p. 103621, 2023. doi: 10.1016/j.jnca.2023.103621
- [26] D. Mimran, R. Bitton, Y. Kfir, E. Klevansky, O. Brodt, H. Lehmann, Y. Elovici, and A. Shabtai, "Security of open radio access networks," *Computers & Security*, vol. 122, p. 102890, 2022. doi: 10.1016/j.cose.2022.102890
- [27] J. Groen, S. D'Oro, U. Demir, L. Bonati, M. Polese, T. Melodia, and K. R. Chowdhury, "Implementing and evaluating security in O-RAN: Interfaces, intelligence, and platforms," *IEEE Network*, vol. 39, no. 1, pp. 227–234, 2025. doi: 10.1109/MNET.2024.3434419
- [28] H. Fard, I. Komarov, and G. Wunder, "Cross-layer intrusion detection in 5G O-RAN: Gains and limits of fusing radio telemetry with network flow records," arXiv:2606.22450, 2026. [Online]. Available: <https://arxiv.org/abs/2606.22450>
- [29] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: Bot-IoT dataset," *Future Generation Computer Systems*, vol. 100, pp. 779–796, 2019. doi: 10.1016/j.future.2019.05.041
- [30] N. Moustafa, "A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets," *Sustainable Cities and Society*, vol. 72, p. 102994, 2021. doi: 10.1016/j.scs.2021.102994
- [31] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, "Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning," *IEEE Access*, vol. 10, pp. 40 281–40 306, 2022. doi: 10.1109/ACCESS.2022.3165809
- [32] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, p. 5941, 2023. doi: 10.3390/s23135941
- [33] I. Sharafaldin, A. Habibi Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. 4th Int. Conf. on Information Systems Security and Privacy (ICISSP)*, 2018, pp. 108–116. doi: 10.5220/0006639801080116
- [34] S. Samarakoon, Y. Siriwardhana, P. Porambage, M. Liyanage, S.-Y. Chang, J. Kim, J. Kim, and M. Ylianttila, "5G-NIDD: A comprehensive network intrusion detection dataset generated over 5G wireless network," *IEEE DataPort*, 2022. doi: 10.21227/xtpe-hv36
- [35] Y. Siriwardhana, S. Samarakoon, P. Porambage, M. Liyanage, S.-Y. Chang, J. Kim, J. Kim, and M. Ylianttila, "Descriptor: 5G wireless network intrusion detection dataset (5G-NIDD)," *IEEE Data Descriptions*, vol. 2, pp. 358–369, 2025. doi: 10.1109/IEEEDATA.2025.3592888
- [36] F. A. Zadeh, A. Civciss, V. Ravihansa, C. Sandeepa, and M. Liyanage, "Descriptor: 5G open radio access network multi-modal intrusion detection dataset (NetsLab-5GORAN-IDD)," *IEEE Data Descriptions*, vol. 2, pp. 348–357, 2025. doi: 10.1109/IEEEDATA.2025.3614167
- [37] M. Sarhan, S. Layeghy, N. Moustafa, and M. Portmann, "NetFlow datasets for machine learning-based network intrusion detection systems," in *Big Data Technologies and Applications (BDTA/WiCon), LNCS*, vol. 371, 2021, pp. 117–135. doi: 10.1007/978-3-030-72802-1_9
- [38] M. Sarhan, S. Layeghy, and M. Portmann, "Towards a standard feature set for network intrusion detection system datasets," *Mobile Networks and Applications*, vol. 27, no. 1, pp. 357–370, 2022. doi: 10.1007/s11036-021-01843-0
- [39] G. Engelen, V. Rimmer, and W. Joosen, "Troubleshooting an intrusion detection dataset: the CICIDS2017 case study," in *Proc. IEEE Security and Privacy Workshops (SPW)*, 2021, pp. 7–12. doi: 10.1109/SPW53761.2021.00009
- [40] R. Flood, G. Engelen, D. Aspinall, and L. Desmet, "Bad design smells in benchmark NIDS datasets," in *Proc. IEEE European Symposium on Security and Privacy (EuroS&P)*, 2024, pp. 658–675. doi: 10.1109/EuroSP60621.2024.00042
- [41] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in *Proc. 31st USENIX Security Symposium*, 2022, pp. 3971–3988. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity22/presentation/arp>
- [42] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023. doi: 10.1016/j.patter.2023.100804
- [43] F. Pendlebury, F. Pierazzi, R. Jordaney, J. Kinder, and L. Cavallaro, "TESSERACT: Eliminating experimental bias in malware classification across space and time," in *Proc. 28th USENIX Security Symposium*, 2019, pp. 729–746.
- [44] G. Apruzzese, P. Laskov, E. Montes de Oca, W. Mallouli, L. Brdalo Rapa, A. V. Grammatopoulos, and F. Di Franco, "The role of machine learning in cybersecurity," *Digital Threats: Research and Practice*, vol. 4, no. 1, pp. 1–38, 2023. doi: 10.1145/3545574
- [45] D. Kus, E. Wagner, J. Pennekamp, K. Wolsing, I. B. Fink, M. Dahlmanns, K. Wehrle, and M. Henze, "A false sense of security? Revisiting the state of machine learning-based industrial intrusion detection," in *Proc. ACM Cyber-Physical System Security Workshop (CPSS)*, 2022, pp. 73–84. doi: 10.1145/3494107.3522773
- [46] L. D'hooge, T. Wauters, B. Volckaert, and F. De Turck, "Inter-dataset generalization strength of supervised machine learning methods for intrusion detection," *Journal of Information Security and Applications*, vol. 54, p. 102564, 2020. doi: 10.1016/j.jisa.2020.102564
- [47] M. Saerens, P. Latinne, and C. Decaestecker, "Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure," *Neural Computation*, vol. 14, no. 1, pp. 21–41, 2002. doi: 10.1162/089976602753284446

- [48] Z. C. Lipton, Y.-X. Wang, and A. J. Smola, "Detecting and correcting for label shift with black box predictors," in *Proc. 35th Int. Conf. on Machine Learning (ICML)*, 2018, pp. 3122–3130.
- [49] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [50] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, and J. Snoek, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.
- [51] T. Fawcett and A. Niculescu-Mizil, "PAV and the ROC convex hull," *Machine Learning*, vol. 68, no. 1, pp. 97–106, 2007. doi: 10.1007/s10994-007-5011-0
- [52] S. Rabanser, S. Günnemann, and Z. C. Lipton, "Failing loudly: An empirical study of methods for detecting dataset shift," in *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.
- [53] B. Sun, J. Feng, and K. Saenko, "Return of frustratingly easy domain adaptation," in *Proc. 30th AAAI Conf. on Artificial Intelligence*, 2016, pp. 2058–2065. doi: 10.1609/aaai.v30i1.10306
- [54] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003. doi: 10.1023/A:1024068626366
- [55] B. Nugraha, M. Hajizadeh, T. Niehoff, A. V. Jnanashree, T. V. Phan, D. Triantafyllopoulou, O. Krause, M. Mieth, K. Moessner, and T. Bauschert, "A comprehensive 5G dataset for control and data plane security and resource management," in *Proc. IEEE International Conference on Cyber Security and Resilience (CSR)*, 2025, pp. 326–333. doi: 10.1109/CSR64739.2025.11130023
- [56] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. doi: 10.1023/A:1010933404324
- [57] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016, pp. 785–794. doi: 10.1145/2939672.2939785
- [58] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: <https://jmlr.org/papers/v12/pedregosa11a.html>
- [59] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *Proc. Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1724–1734. doi: 10.3115/v1/D14-1179
- [60] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. on Data Mining (ICDM)*, 2008, pp. 413–422. doi: 10.1109/ICDM.2008.17
- [61] X. Jin, L. E. Li, L. Vanbever, and J. Rexford, "SoftCell: Scalable and flexible cellular core network architecture," in *Proc. 9th ACM Conf. on Emerging Networking Experiments and Technologies (CoNEXT)*, 2013, pp. 163–174. doi: 10.1145/2535372.2535377
- [62] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017, pp. 4765–4774. [Online]. Available: <https://papers.nips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>



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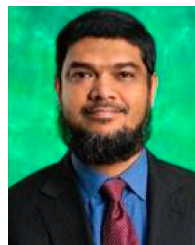


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cyber-physical systems.

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